
TraceDx: Criticality-Weighted Atomic Facts as a Training Signal for Sequential Clinical Diagnosis

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Abstract

1 In clinical decision-making, success depends on taking the right action to obtain
2 the evidence that matters. One targeted clinical workup can identify a diagnosis
3 that a battery of nonspecific tests missed. Large language models have shown
4 considerable promise in clinical diagnosis, but most current approaches assume
5 that all relevant patient information is available upfront, which does not reflect how
6 diagnostic evidence is gathered in practice. Even when models gather evidence
7 iteratively, challenges arise in determining which action to take, as they receive
8 no feedback on which findings are diagnostically decisive. To solve this problem,
9 we introduce TraceDx, a reinforcement learning framework for sequential clinical
10 diagnosis. TraceDx is built on the idea that unstructured clinical records can be
11 decomposed into atomic clinical findings, each scored by its diagnostic value. This
12 signal rewards agents for discovering decisive evidence in addition to reaching
13 the correct diagnosis. Consistent with this design, a case-level analysis shows
14 that recovery of critical evidence is the only statistically significant predictor of
15 diagnostic success. Direct case comparisons further show that TraceDx uncovers
16 more diagnostically important evidence, and uses it to resolve the key diagnostic
17 uncertainty. TraceDx-trained open-weight models outperform multiple frontier
18 baselines on both MIMIC-CDM and a rare gastrointestinal disease dataset.

19 1 Introduction

20 Clinical diagnosis is inherently a sequential process in which evidence is acquired through questions
21 asked, tests ordered, and results interpreted [4]. However, most approaches to clinical diagnosis with
22 Large Language Models (LLMs) reduce this process to inference over a fixed input [21, 15, 16, 9, 5, 3].
23 This approach assumes all pertinent information is available upfront and overlooks how evidence
24 is gathered iteratively in real clinical practice. A few benchmarks let agents acquire information
25 iteratively [17, 12, 6], but they rely on LLMs without task-specific training.

26 This raises the question of how to train models to operate in clinical settings, wherein diagnosis
27 depends on gathering the right evidence through a sequence of actions. The challenge is that not
28 all clinical evidence is equally informative. Some findings are decisive, such as a CT confirming
29 diverticulitis or a markedly elevated lipase indicating pancreatitis, whereas other findings provide
30 only weak diagnostic signals or none at all. Clinicians succeed in part by prioritizing the evidence that
31 best distinguishes among competing hypotheses, selectively pursuing tests that resolve uncertainty,
32 rather than gathering information indiscriminately [18, 24, 2].

33 Training an agent to recognize these differences in diagnostic importance requires feedback on what
34 its actions actually reveal, since the value of a diagnostic action lies in the evidence it uncovers.
35 Yet such feedback is hard to obtain. The patient record presents an unstructured stream of notes,
36 test results, and observations, mixing raw findings with clinical interpretations layered on top. At

37 this granularity, the record offers no scorable unit. The findings are there, but not as discrete items
38 the agent can be credited for recovering. We address this by decomposing each patient record into
39 atomic clinical facts, drawing on prior work in factuality evaluation [14]. Each atomic fact is a single
40 observation, a lab value, an imaging finding, or a physical exam result.

41 Atomization gives us scorable units, but no way to weigh them. Prior factuality work treats extracted
42 facts as equally weighted and uses them to evaluate model outputs. In clinical decision making, equal
43 weighting fails because findings differ in how much they narrow the diagnosis. We score each fact
44 on an ordinal scale by its diagnostic criticality. The scale runs from clinically irrelevant evidence to
45 pathognomonic findings that settle the diagnosis. Atomization and criticality scoring give us a way
46 to value each piece of evidence. The remaining problem is how to leverage these per-fact values to
47 assess action quality across a trajectory.

48 We propose TraceDx, a reinforcement learning framework that turns these per-fact values into a
49 trajectory-level reward, crediting agents for the value of the evidence their actions discover. A
50 trajectory that uncovers more decisive evidence receives higher credit, even for the same diagnosis.
51 The agent operates in an unconstrained clinical simulator, where actions and diagnoses are proposed
52 in free text with no predefined menu. In response, the simulator generates observations grounded in
53 the patient record. The simulator environment mirrors what a clinician faces in practice, leaving the
54 agent to decide which tests are worth pursuing. The goal is to train an agent that learns not just which
55 diagnosis to reach, but which actions to take and why.

56 We evaluate TraceDx on two clinical diagnostic datasets. MIMIC-CDM [6] contains real emergency
57 department cases derived from MIMIC-IV, spanning common abdominal pathologies. The rare
58 disease dataset [10] draws from published medical case reports covering uncommon gastrointestinal
59 and hepatobiliary conditions. The two settings present different challenges. In MIMIC-CDM, the
60 agent must isolate a handful of decisive findings within hundreds of routine measurements per case,
61 while in the rare disease set it must recognize the hallmark of a condition it has barely encountered.
62 Across both settings, TraceDx-trained open-weight models match or exceed most frontier baselines,
63 despite operating at a smaller scale. A case-level analysis identifies recovery of critical evidence as
64 the only significant predictor of diagnostic success, and case studies show TraceDx-trained agents
65 resolving diagnostic uncertainty with more decisive actions.

66 2 Related Work

67 **Clinical diagnosis with LLMs.** Prior work on LLMs for clinical diagnosis has largely operated
68 in a non-interactive setting where the model receives a complete case and produces a diagnosis or
69 differential. This includes multiple-choice medical QA [21, 22] and complex case vignettes drawn
70 from published case reports [5, 3]. Several efforts move toward interaction, where the model must
71 request evidence before committing to a diagnosis. Hager et al. [6] introduce MIMIC-CDM, an
72 interactive diagnostic benchmark built on MIMIC-IV. Subsequent benchmarks evaluate stepwise
73 evidence acquisition on clinical case vignettes [17, 12] and across multiple specialties [19]. All of
74 these efforts evaluate fixed models rather than training them on specific tasks.

75 A smaller body of work addresses optimization. AMIE [28] trains a conversational agent for
76 history-taking via supervised fine-tuning on self-play dialogues with a simulated patient. Yu et
77 al. [31] train policies for cost-effective sequential test ordering over tabular lab values, and ED-
78 Copilot [25] demonstrates LM-guided diagnostic ordering in emergency settings. A concurrent
79 effort, LA-CDM [1], applies SFT and GRPO to a two-agent system on MIMIC-CDM, with a sparse
80 terminal correctness reward augmented by per-test cost penalties, plus a separate GRPO objective for
81 confidence calibration. LA-CDM operates over a predefined menu of four abdominal diagnoses and
82 twelve test types. TraceDx differs in two ways. First, it trains a single-agent policy with a reward that
83 quantifies the diagnostic value of discovered evidence, rewarding evidence quality rather than just
84 terminal correctness. Second, it operates in an unconstrained simulator where actions and diagnoses
85 are proposed in natural language, supporting open-ended exploration over the full action space. These
86 choices let TraceDx learn what evidence matters, not just what diagnosis to output.

87 **Multi-turn RL for LLM agents.** Search-R1 [8] trains LLM agents with a binary terminal reward on
88 the final answer. TraceDx replaces this with a richer terminal reward that credits trajectories not only
89 for correctness but for discovering diagnostically decisive evidence.

Fact decomposition. Fact decomposition is the first step in the decompose-then-verify pipeline for long-form factuality evaluation introduced by FActScore [14] and extended in subsequent work [29, 23]. MedScore [7] adapts this paradigm to free-form medical text by introducing condition-aware claim decomposition for verification against medical corpora. While some work uses decompose-then-verify scores as training signals [27, 13], these efforts decompose model outputs into atomic facts and use the resulting score as a preference signal for offline alignment of a generator LLM. All these methods treat extracted facts as equally informative, which fails in diagnostic reasoning where findings differ in how much they narrow the diagnosis. Like these methods, we apply LLM-based decomposition to extract atomic facts. We depart from this paradigm in three ways. We decompose ground-truth patient records rather than model outputs, weight each fact by its diagnostic importance, and use the resulting signal as an online RL reward to train a diagnostic agent.

3 Preliminaries

Sequential Diagnostic Reasoning. We frame medical diagnosis as a sequential decision-making problem and model it as a finite-horizon POMDP in which the hidden state is the complete patient record M , fixed throughout the episode. An agent π_θ begins with an initial patient presentation o_0 and iteratively gathers further clinical evidence through a series of actions a_t , each yielding a clinical observation

$$o_{t+1} \sim \mu(\cdot \mid M, \tau_t, a_t),$$

where μ is the clinical simulator (Section 4.1), and $\tau_t = (o_0, a_0, o_1, \dots, a_{t-1}, o_t)$ is the interaction history up to time t . Since M is never directly observed, the agent takes actions based on the history, yielding a history-dependent policy $\pi_\theta(a_t \mid \tau_t)$. Each episode terminates when the agent commits to a final diagnosis $\hat{d}$ or reaches the maximum horizon $T_{\max}$. The resulting trajectory $\tau = (o_0, a_0, o_1, a_1, \dots, a_{T-1}, o_T, \hat{d})$ records the full diagnostic process. Here $T \leq T_{\max}$ is the number of actions taken. If the horizon is reached without a committed diagnosis, $T = T_{\max}$ and $\hat{d} = \emptyset$.

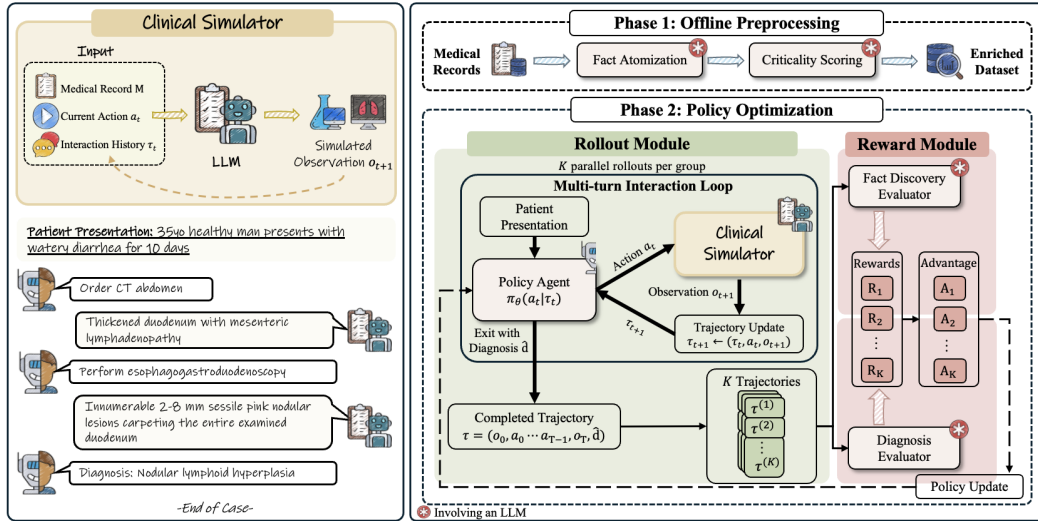


Figure 1: Overview of the TraceDx framework. *Left:* The agent issues a diagnostic action a_t at each step and the simulator returns an observation based on the patient record M and interaction history τ_t . *Right:* Two-phase training pipeline. Phase 1 extracts atomic clinical facts from each record offline and scores them by diagnostic importance. Phase 2 trains the agent with the criticality-weighted reward $R(\tau)$.

114 4 Method

115 We propose TraceDx, a framework that turns fact-level criticality into a training signal, rewarding
 116 agents for discovering the facts that actually drive diagnosis rather than merely for arriving at the
 117 right answer. TraceDx consists of four components: (1) an unconstrained clinical simulator for
 118 multi-step diagnostic interaction, (2) an offline fact atomization and criticality scoring pipeline, (3) a
 119 criticality-weighted reward, and (4) policy optimization.

120 4.1 Clinical Simulator Setup

121 The simulator μ is implemented as an LLM grounded in the patient record M . Both the action a_t and
 122 the eventual diagnosis $\hat{d}$ are produced in free-text form, without a predefined action set or diagnosis
 123 label space. This design captures the open-ended nature of clinical workups, where clinicians choose
 124 tests and diagnoses freely rather than from a fixed menu. The gold diagnosis d^* is used solely for
 125 reward computation and is never provided to the agent or simulator. Providing d^* to the simulator
 126 would risk generating confirmatory observations that bias the trajectory toward the correct answer,
 127 introducing a signal that would not be available in real clinical practice. An example conversation
 128 between the agent and the simulator is shown in the left panel of Figure 1. Further implementation
 129 details, limitation analysis, and validation results are in Appendix C.

130 4.2 Fact Atomization and Criticality Scoring

131 The patient record is unstructured and heterogeneous, mixing raw findings with clinical interpretations.
 132 To obtain a fine-grained training signal, we decompose patient records into atomic clinical facts and
 133 assign each fact a criticality score reflecting its diagnostic importance. This produces, for every case,
 134 a set of scorable units annotated with their relative diagnostic value (Figure 2). These scores are used
 135 in Section 4.3 to construct a trajectory-level reward.

136 Before training, we preprocess each patient record M in the dataset to extract a set of atomic clinical
 137 facts. Formally, we define a fact extraction function ϕ such that $F = \phi(M) = \{f_1, \dots, f_n\}$, where
 138 each f_i is an atomic clinical finding, a single observation such as a lab value, imaging finding, or
 139 physical examination result, stated without interpretation or inference.

140 Each fact is then scored by its diagnostic importance. Given the fully extracted fact set F , gold
 141 diagnosis d^* , and initial patient presentation o_0 , an LLM-based scorer assigns each fact a criticality
 142 score $c(f) \in \{0, 1, 2, 3\}$ conditioned on d^* and o_0 . Higher scores indicate greater clinical importance:
 143 $c = 0$ for clinically irrelevant evidence, $c = 1$ for supportive but non-specific evidence, $c = 2$ for
 144 strongly supportive evidence, and $c = 3$ for pathognomonic or hallmark findings. This scoring is
 145 performed offline prior to training, adding no computational overhead during rollout. Full details are
 in Appendix D.

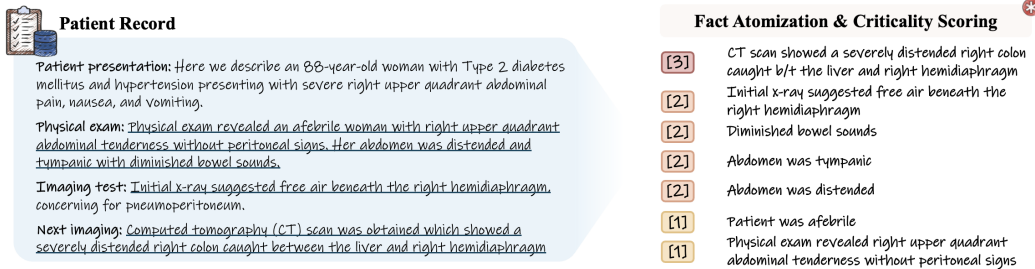


Figure 2: Fact atomization and criticality scoring. A patient record is decomposed into atomic findings, each scored by diagnostic importance $c(f) \in \{0, 1, 2, 3\}$. Underlined text marks content extracted as atomic facts. Initial patient presentation and clinical interpretations are excluded.

4.3 Criticality-Weighted Reward

A naive baseline is to optimize the policy using a binary outcome-based reward that depends only on the final diagnostic accuracy: $R_{\text{naive}}(\tau) = \text{acc}(\hat{d}, d^*)$, where $\text{acc}(\hat{d}, d^*) \in \{0, 1\}$ is evaluated by an LLM judge assessing clinical equivalence rather than exact string match. Diagnostic reasoning involves both gathering evidence and reasoning over it, but terminal diagnostic accuracy supervises only the latter, leaving the agent without a signal to learn the former.

To address this limitation, we design a mechanism to measure how much diagnostically relevant information is uncovered along a trajectory, which serves as a proxy for the quality of the actions taken. Given the fact set F and a completed trajectory τ , we let $D(\tau) \subseteq F$ denote the subset of facts revealed during the trajectory, computed by an LLM-based evaluator that parses the trajectory and outputs indices of observed facts (see Appendix E.1).

We quantify the amount of diagnostically important evidence recovered using the criticality recall:

$$CR(\tau) = \frac{\sum_{f \in D(\tau)} c(f)}{\sum_{f \in F} c(f)}.$$

We restrict training to cases with $\sum_{f \in F} c(f) > 0$ so that $CR(\tau)$ is well-defined.

We define the criticality-weighted reward as

$$R(\tau) = (\alpha \cdot CR(\tau) + \beta) \text{acc}(\hat{d}, d^*) - \lambda \frac{T}{T_{\max}},$$

where $\alpha, \lambda \geq 0$ and $\beta > 0$. Trajectories that uncover more diagnostically decisive facts receive higher reward for the same correct diagnosis, while the positive offset β ensures that correct diagnoses remain rewarded even when criticality recall is low. The step penalty $-\lambda T/T_{\max}$ discourages unnecessary actions and encourages efficient reasoning.

The following proposition establishes two guarantees under approximate calibration: the CR component orders trajectories by true evidence quality whenever scoring noise is small, and the full reward inherits this ordering for correct trajectories whenever their CR gap exceeds the step-penalty difference.

Proposition 1 (Reward order preservation under approximate calibration). *Assume $\sum_{f \in F} c(f) > 0$. Let each fact $f \in F$ have latent relevance $r_f \geq 0$, and suppose the scorer satisfies*

$$c(f) = \kappa r_f + \varepsilon_f, \quad \kappa > 0, \quad |\varepsilon_f| \leq \sigma,$$

where $\sigma \geq 0$ is the worst-case per-fact scoring error. For two trajectories $\tau^{(1)}, \tau^{(2)}$ from the same patient case (sharing fact set F) with discovered sets $D_i = D(\tau^{(i)})$, define $\nu(D) = \sum_{f \in D} r_f$. If

$$\nu(D_1) - \nu(D_2) > \frac{\sigma}{\kappa} (|D_1| + |D_2|),$$

then $CR(\tau^{(1)}) > CR(\tau^{(2)})$. Furthermore, if $\alpha > 0$, both trajectories are correct, and $CR(\tau^{(1)}) - CR(\tau^{(2)}) > \frac{\lambda}{\alpha} \cdot \frac{T_1 - T_2}{T_{\max}}$, where T_i is the number of actions in $\tau^{(i)}$, then $R(\tau^{(1)}) > R(\tau^{(2)})$.

In short, the criticality recall ranks trajectories by true evidence quality whenever scoring errors are small relative to the true relevance gap. The full reward inherits this ranking for correct trajectories whenever their CR gap exceeds the step-penalty difference. A formal proof is provided in Appendix H, and validation of the criticality scorer is detailed in Appendix D.

4.4 Policy Optimization

We optimize the policy π_θ using an importance-weighted REINFORCE objective with GRPO-style group-relative advantages [20], omitting PPO-style clipping and the KL penalty. For each patient case, we sample a group of K trajectories $\{\tau^{(1)}, \dots, \tau^{(K)}\}$, each receiving a scalar terminal reward $R_k = R(\tau^{(k)})$. We form the group-relative advantage $A_k = R_k - \text{mean}(R_1, \dots, R_K)$.

Each clinical action a_t is realized as a sequence of agent-generated tokens, so we apply the policy gradient at the token level. Our training objective is:

$$\mathcal{L}(\theta) = - \sum_{k=1}^K \sum_l m_{k,l} \rho_{k,l} A_k,$$

where $y_{k,l}$ is the l -th token of trajectory $\tau^{(k)}$, $y_{k,<l}$ denotes the preceding tokens in $\tau^{(k)}$, $\rho_{k,l} = \pi_\theta(y_{k,l} \mid y_{k,<l}) / \pi_{\text{old}}(y_{k,l} \mid y_{k,<l})$ is the per-token importance ratio, and $m_{k,l} \in \{0, 1\}$ is 1 for agent-generated tokens and 0 otherwise. We mask non-agent tokens by zeroing their per-token advantage, analogous to retrieved-token masking in Search-R1 [8]. π_{old} denotes the policy used to generate the rollouts, whose log-probabilities are recorded at sampling time and held fixed during the gradient update. Groups with identical rewards across all K trajectories yield $A_k = 0$ and are excluded.

5 Experiments

5.1 Experiment Setup

Datasets. We evaluate our method on two clinical diagnostic datasets: *MIMIC-CDM* and a *Rare Disease* dataset. We sample 250 cases from MIMIC-CDM [6], focusing on three conditions with distinct diagnostic structures: diverticulitis, cholecystitis, and pancreatitis. To ensure reliable evaluation with an LLM-based grader, we restrict to cases with short, well-defined diagnoses.

The Rare Disease dataset [10] consists of 268 cases drawn from published medical case reports, spanning a wide range of uncommon conditions. We report results on held-out test sets of 75 MIMIC-CDM cases and 81 Rare Disease cases. Detailed dataset construction and filtering criteria are provided in Appendix B.

Baselines. We compare against closed-weight frontier LLMs (GPT-5, GPT-5.1, GPT-5.4, Sonnet 4, Sonnet 4.5, Opus 4.7, Gemini 2.5 Pro, Gemini 3.1 Pro) and the untrained base models Qwen3-32B [30] and Kimi-K2.5 [26]. We also include GPT-5.1 with oracle context as an approximate upper bound, where the full patient record is provided upfront, removing the evidence-acquisition burden and bounding what any interactive agent could achieve with complete information. Further implementation details for all methods are in Appendices F and G. We additionally compare against physicians playing the same simulator game on a smaller subset of the held-out test set (see Appendix J).

Evaluation. Models are evaluated on diagnostic accuracy via interaction with the clinical simulator, with Claude Sonnet 4 as both the simulator backbone and the diagnosis grader. The simulator, criticality scorer, and diagnosis grader were each validated against physician judgment (Appendices C.3, D.3, and E.3).

5.2 Accuracy Analysis

Table 1: Diagnostic accuracy across Rare Disease and MIMIC-CDM datasets. **Bold**: best frontier. Underline: best open-weight.

	Model	Rare Disease	MIMIC-CDM
Frontier LLMs	GPT-5	0.419 $\pm$ 0.013	0.567 $\pm$ 0.016
	GPT-5.1	0.407 $\pm$ 0.025	0.577 $\pm$ 0.040
	GPT-5.4	0.374 $\pm$ 0.025	0.604 $\pm$ 0.046
	Sonnet 4	0.319 $\pm$ 0.019	0.519 $\pm$ 0.026
	Sonnet 4.5	0.357 $\pm$ 0.040	0.595 $\pm$ 0.040
	Opus 4.7	0.456 $\pm$ 0.049	0.653 $\pm$ 0.073
	Gemini 2.5 Pro	0.386 $\pm$ 0.047	0.573 $\pm$ 0.038
	Gemini 3.1 Pro	0.435 $\pm$ 0.014	0.560 $\pm$ 0.013
Qwen3-32B	Base	0.316 $\pm$ 0.028	0.555 $\pm$ 0.024
	TraceDx	0.370 $\pm$ 0.024	0.600 $\pm$ 0.027
Kimi-K2.5	Base	0.378 $\pm$ 0.026	0.568 $\pm$ 0.028
	TraceDx	<u>0.424 $\pm$ 0.026</u>	<u>0.613 $\pm$ 0.026</u>
Oracle	GPT-5.1 (full context)	0.543 $\pm$ 0.032	0.680 $\pm$ 0.013

Table 1 reports diagnostic accuracy across both datasets. TraceDx delivers consistent improvements over the untrained base in every setting we tested. Both base models gain roughly 4.5 to 5.4 percentage points across the two datasets. The gains are nearly identical across base models of different scale and architecture, and across datasets with very different fact distributions. The consistency across

architectures and datasets suggests the gains are attributable to the TraceDx framework rather than to the capabilities of any particular base model.

Beyond improving over the base models, TraceDx makes open-weight models competitive with frontier models, indicating that training signal, not scale alone, drives diagnostic performance in this setting. Kimi-K2.5 with TraceDx surpasses most frontier models on both datasets, ranking second on MIMIC-CDM and within the top three on Rare Disease. Qwen3-32B with TraceDx also surpasses most frontier models on MIMIC-CDM, despite its base trailing the majority of them. The remaining gap to the oracle baseline reflects the cost of acquiring evidence interactively.

5.3 Ablation Study

We conduct an ablation study to identify the source of TraceDx’s gains. We compare TraceDx against an untrained Base and a Diagnosis-only (Dx) ablation that swaps TraceDx’s criticality reward for binary correctness. This comparison separates the effect of reinforcement learning itself from that of explicitly training for evidence acquisition. Diagnosis-only fails to improve over the base model and even slightly degrades performance on MIMIC-CDM, whereas TraceDx performs best on both datasets. The results show that TraceDx’s gains come from its criticality-weighted reward rather than reinforcement learning alone. We next test the premise underlying the criticality-weighted reward, asking whether evidence quality predicts diagnostic success.

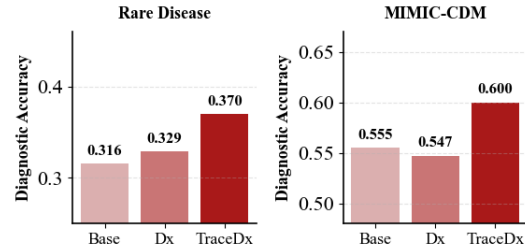


Figure 3: Ablation for Qwen3-32B.

5.4 Evidence Quality Predicts Diagnostic Success

We focus this analysis on the rare disease data, where criticality annotations are well-distributed across all four levels (0–3), enabling reliable discrimination between facts of different diagnostic value. In MIMIC-CDM, more than 80% of facts receive criticality 0, which compresses the signal range and makes case-level estimates of hallmark-evidence concentration too noisy for inference (Figure 7; Appendix D.2). The rare disease set therefore provides a cleaner setting for testing whether evidence quality predicts diagnostic success.

To test this premise, each of the twelve models in Table 1 first plays the simulator game on rare-disease cases to gather its own evidence. We then pass that evidence to a fixed GPT-5.1 reasoner that produces the diagnosis. Holding the reasoner constant ensures that any variation in diagnostic success reflects evidence quality, not diagnostic reasoning capability. We characterize each evidence set along three dimensions: *Crit-3 ratio*, the fraction of retrieved facts with $c(f) = 3$; *noise ratio*, the fraction of retrieved facts with $c(f) = 0$; and *coverage*, the total fraction of available facts retrieved. We fit a logistic regression predicting diagnostic success from these three features, with agent fixed effects and standard errors clustered by case.

Table 2: Logistic regression of diagnostic success on evidence features (N=972, agent fixed effects).

Predictor	Coef.	<i>p</i>
Crit-3 Ratio	6.61	0.002
Noise Ratio	−0.16	0.803
Coverage	0.11	0.862
Agent Fixed Effects	Included	

Crit-3 ratio is the only significant predictor of diagnostic success, while noise ratio and coverage are both non-significant (Table 2). The fixed effects ensure this reflects within-agent variation in evidence quality. Holding model identity constant, hallmark concentration is what separates correct from incorrect diagnoses. The criticality reward is designed to shift evidence acquisition toward hallmark findings, and the regression shows that hallmark concentration is the strongest predictor

of diagnostic success. These findings are consistent with TraceDx’s premise that diagnostic success depends on evidence quality, not quantity.

5.5 Performance by Disease Category

Overall accuracy can obscure important differences in how models behave across diseases with distinct diagnostic structures. We therefore break down performance on MIMIC-CDM by disease category to ask when criticality-guided training helps most.

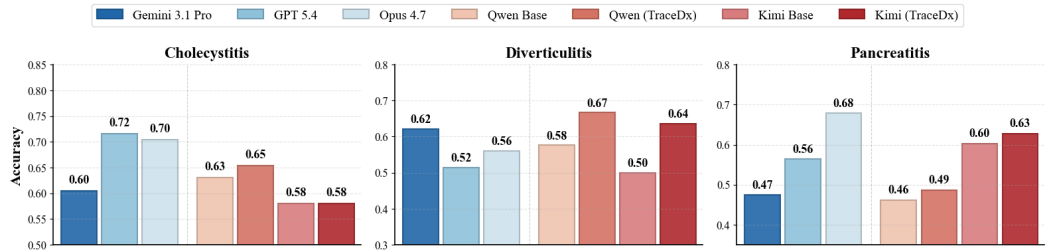


Figure 4: Mean diagnostic accuracy per disease category across models.

Cholecystitis. TraceDx yields a modest improvement for Qwen3-32B but no gain for Kimi-K2.5. Qwen3-32B (TraceDx) surpasses Gemini 3.1 Pro but trails GPT 5.4 and Opus 4.7. Cholecystitis has no single pathognomonic finding. Diagnosis requires satisfying criteria across local signs, systemic markers, and imaging, which may dilute the benefit of criticality-guided training.

Diverticulitis. TraceDx shows the clearest gains here, with both models improving substantially over their respective base models and surpassing all three frontier models shown. This is consistent with the relatively direct diagnostic structure of diverticulitis, where CT imaging provides a clear and decisive signal.

Pancreatitis. Both TraceDx models improve modestly over their base models by a similar margin. Kimi-K2.5 with TraceDx surpasses two of the three frontier models shown (Gemini 3.1 Pro and GPT 5.4), trailing only Opus 4.7. Qwen3-32B Base starts substantially weaker than Kimi-K2.5 Base on pancreatitis, which explains why the TraceDx variants diverge. The gap reflects the base models’ relative strengths rather than a differential effect of criticality-guided training.

TraceDx shows the clearest gains on diverticulitis, where both models exceed frontier performance, and limited benefit on cholecystitis. On pancreatitis, both models yield consistent modest gains, with the divergence between the two TraceDx models explained by their starting points.

5.6 Case Study

We examine one case from each dataset to illustrate how TraceDx shapes diagnostic reasoning.

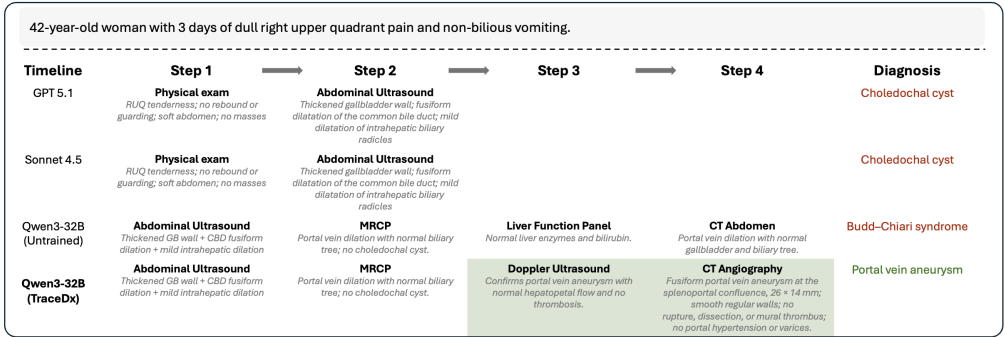


Figure 5: Diagnostic trajectories for portal vein aneurysm. The grey box shows the initial presentation. Each row shows a model’s sequence of diagnostic actions and the resulting final diagnosis. Dark green boxes mark the actions that resolve the key diagnostic uncertainty.

287 **Case 1 (Portal Vein Aneurysm)**

288 A 42-year-old woman presented with right upper quadrant pain and vomiting (Figure 5). Initial ultra-
289 sound showed a dilated common bile duct and intrahepatic biliary radicles, suggesting a choledochal
290 cyst, a congenital bile-duct abnormality. GPT-5.1 and Sonnet 4.5 accepted this framing and concluded
291 choledochal cyst without further workup. Qwen3-32B Base obtained an MRCP that revealed a normal
292 biliary tree but a dilated portal vein, directly contradicting the bile-duct interpretation, but misread
293 this as Budd-Chiari syndrome, a different vascular disorder. TraceDx recognized the contradiction,
294 shifted its hypothesis from biliary to vascular pathology, ordered a Doppler ultrasound confirming a
295 portal vein aneurysm, and used CT angiography to characterize the lesion. TraceDx succeeds not by
296 gathering more evidence, but by being willing to abandon a working hypothesis when the evidence
contradicts it.

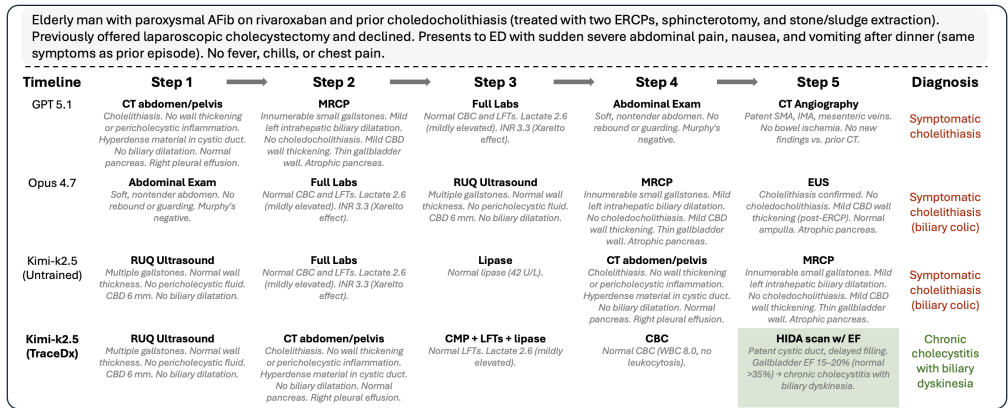


Figure 6: Diagnostic trajectories for chronic cholecystitis. Same layout as Figure 5.

297

298 **Case 2 (Chronic Cholecystitis)**

299 An elderly man with recurrent biliary pain posed a different challenge (Figure 6). Standard tests came
300 back negative for acute pathology across all models, narrowing the differential to two conditions.
301 Symptomatic cholelithiasis refers to gallstones causing pain in a structurally normal gallbladder.
302 Chronic cholecystitis with biliary dyskinesia presents with the same imaging appearance but reflects
303 a gallbladder that has lost its ability to contract. The two are indistinguishable on ultrasound and CT,
304 and the only differentiating test is a HIDA scan, which measures gallbladder contraction directly.
305 The baselines, including Opus 4.7, stopped at symptomatic cholelithiasis without testing gallbladder
306 function. TraceDx ordered the HIDA scan, and the resulting ejection fraction of 15–20%, well below
307 the normal threshold of 35%, established chronic cholecystitis with biliary dyskinesia. The case
308 shows that TraceDx improves not only by selecting decisive actions, but by recognizing when a
309 standard workup leaves a diagnostically pivotal question unresolved.

310 **6 Conclusion**

311 TraceDx trains diagnostic agents on the evidence they recover, not just the diagnoses they reach. We
312 decompose each patient record into atomic clinical facts and score them by diagnostic criticality,
313 building a fine-grained training signal where prior approaches saw only a final outcome. Agents are
314 rewarded for recovering decisive evidence in an open-ended simulator with no predefined actions or
315 diagnoses, training them toward evidence quality rather than terminal accuracy. A regression shows
316 hallmark concentration as the only significant predictor of diagnostic success, and an ablation traces
317 the gain to the criticality-weighted reward. Across two datasets, TraceDx-trained open-weight models
318 match or exceed multiple frontier baselines. Looking ahead, this fact-atomization and criticality-
319 scoring pipeline could serve as a reusable training and evaluation signal in other expert domains
320 where decisions hinge on evidence quality.

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B Dataset Details

B.1 MIMIC-CDM Dataset

Data source. Cases are drawn from MIMIC-IV [6], a large de-identified electronic health record database. The source pool contains 648 cholecystitis, 257 diverticulitis, and 538 pancreatitis admissions (1,443 total). Each admission includes patient history (free text), physical examination findings, laboratory tests, microbiology results, radiology reports, and discharge diagnosis.

Dataset fields. After processing, each case in the MIMIC-CDM dataset contains the following fields:

- **case_id:** hospital admission identifier from MIMIC-IV.

- **context**: the full patient record from MIMIC-IV that the simulator uses to ground its responses, containing patient history (free-text narrative of presenting complaint and medical, surgical, social, and family history), physical examination (vital signs and exam findings), laboratory tests (mapped from numeric item IDs to human-readable labels), microbiology results (same mapping), and radiology notes (each with modality, region, and exam name). De-identification in MIMIC-IV replaces identifiers such as age, dates, and names with placeholder tokens, preserved as-is.
- **initial_prompt**: the opening patient presentation provided to the agent at the start of each episode, derived from the patient history component of the context.
- **ground_truth**: the discharge diagnosis, used as the target label for evaluation.
- **source**: disease category label (cholecystitis, diverticulitis, or pancreatitis), used for stratified sampling and per-disease analysis.
- **facts, fact_criticalness**: atomic clinical facts and their diagnostic importance scores; described in Appendix D.
- **procedures_discharge**: discharge procedures associated with the admission.

Disease selection. The original MIMIC-CDM benchmark includes four abdominal conditions; we exclude appendicitis. Hager et al. [6] report that appendicitis achieves the highest LLM diagnostic accuracy among all four conditions, and attribute this to 82.7% of radiologist reports explicitly stating that the appendix is dilated, enlarged, or fluid-filled—reducing the task to pattern recognition rather than multi-signal reasoning. We focus on the three remaining conditions, which require integrating heterogeneous evidence and are more prone to mutual confusion, making them a more demanding diagnostic reasoning benchmark.

Short-answer filter. To enable reliable LLM judge evaluation, we retain only admissions whose discharge diagnosis contains at most 8 words, filtering the majority of patients that have more than two diagnoses. Long discharge summaries tend to be compound or qualified, making LLM-based grading unreliable.

Sampling and splits. After filtering, we sample 100 cases uniformly at random from each disease stratum (`random_state=42`), yielding 300 balanced cases. We then apply a stratified 70/30 train/test split per disease. To reduce the computational cost of running full multi-turn simulation trajectories, we independently subsample 175 cases from the training partition and 75 from the validation partition (`seed=42`), yielding a final dataset of 250 cases while preserving the 70/30 ratio. The final composition is shown in Table 3. Train and test splits are non-overlapping, enforced by assertion.

Table 3: MIMIC-CDM dataset statistics. Source pool counts reflect primary-diagnosis admissions before filtering. Train and test counts reflect the final 250-case subset after the short-answer filter, stratified sampling, and 70/30 split.

Disease	Source Pool	Train	Test
Cholecystitis	648	58	27
Diverticulitis	257	62	22
Pancreatitis	538	55	26
Total	1,443	175	75

B.2 Rare Disease Dataset

Data source. The dataset comprises 268 cases drawn from published gastrointestinal and hepatobiliary case reports in the clinical literature [10].

Dataset fields. Each case contains the following fields:

- **case_id**: DOI-encoded string identifying the source case report.
- **context**: a structured text document with labeled sections reflecting the sequential disclosure of clinical information as it appears in a case report.
- **initial_prompt**: the opening patient presentation provided to the agent at the start of each episode, derived from the *patient_presentation* section. Mirrors the MIMIC-CDM design.

- **ground_truth**: the diagnosis from the original case report, used as the target label for evaluation.
 - **COMMON_UNCOMMON**: rarity label (COMMON or UNCOMMON), used for stratified splitting and per-rarity analysis.
 - **facts, fact_criticalness**: atomic clinical facts and their diagnostic importance scores; described in Appendix D.
- Disease rarity classification.** Each case is labeled COMMON (66 cases, 24.6%) or UNCOMMON (202 cases, 75.4%). UNCOMMON cases correspond to rare diagnoses and form the majority of this dataset, making it substantially more challenging than MIMIC-CDM cases.
- Splits.** All 268 cases are assigned to train or test partitions with no subsampling. Splits are stratified by the COMMON_UNCOMMON label. The final split contains 187 training cases (46 COMMON, 141 UNCOMMON) and 81 test cases (20 COMMON, 61 UNCOMMON).

C Clinical Simulator Details

C.1 Implementation Details

The simulator is implemented using Claude Sonnet 4 as the backbone LLM. It receives the full patient record and the accumulated action–result history as context, and generates a grounded clinical observation in response to each agent action. Separate system prompts are used for the rare disease and MIMIC-CDM datasets to account for dataset-specific formatting and de-identification requirements; both are provided in Appendix L.1.

C.2 Simulator Limitations

A recurring source of stochasticity arises when the agent orders a test not documented in the patient record. When the requested test is absent from the patient record, the simulator generates a plausible result consistent with the patient’s presentation and past actions. Since no ground truth value exists for that test, the generated result varies across independent runs.

We illustrate this with a rare disease case: a 22-year-old man presenting after elemental mercury ingestion. The patient record documents vital signs, routine labs, abdominal imaging, and a single mercury measurement: 24-hour urine mercury of 44 µg/L. No serum or blood mercury value was recorded. Across 7 independent runs, the agent ordered serum or blood mercury, a test not present in the record. Table 4 shows a representative sample of the simulator’s responses.

Table 4: Selected simulator responses across 7 runs in which the agent ordered serum or blood mercury for the same case. Simulated values span two orders of magnitude, units are inconsistent across runs, and in one run the simulator returns the urine value from the record relabeled as a serum result.

Simulated value	Simulator interpretation
(none)	Within normal limits
2.3 mcg/L	Within normal limits; no acute toxicity
12 µg/L	Mildly elevated; consistent with poor GI absorption
15 ng/mL	Mildly elevated (<i>same number as below; different unit</i>)
44 µg/L	Elevated (<i>urine value from record, relabeled as serum</i>)
125 ng/mL	Markedly elevated; confirms acute exposure
185 ng/mL	Markedly elevated; immediate intervention required

A second case illustrates a related but distinct failure pattern: for the same absent test, the simulator’s behavior is inconsistent across runs, sometimes declining to return a result and sometimes simulating one.

The case is a MIMIC-CDM admission with a gold diagnosis of gallstone pancreatitis with choledocholithiasis. The patient record contains an MRCP but no CT with contrast. When the agent ordered CT abdomen with contrast, the simulator refused in one of four runs and simulated a result in the remaining three. Table 5 shows the four runs.

Table 5: Four runs in which the agent ordered CT abdomen with contrast for the same MIMIC-CDM case. In the two runs that simulate without reporting gallstones (rows 2–3), the agent reaches an incorrect diagnosis; the run that simulates inflammatory changes without explicitly excluding gallstones (row 4) produces a correct diagnosis.

Behavior	Simulated CT finding
Declined	CT not present in record
Simulated	Gallbladder wall thickening, CBD dilation $\sim 7\text{--}8$ mm, PD dilation $\sim 3\text{--}4$ mm; <i>no gallstones or choledocholithiasis identified</i>
Simulated	Pancreatic parenchyma with decreased enhancement, PD dilation; <i>no gallstones identified</i>
Simulated	Gallbladder wall thickening with inflammation, CBD $\sim 7\text{--}8$ mm, PD $\sim 3\text{--}4$ mm, pancreatic inflammatory changes

Together, these two cases illustrate the main sources of stochasticity when the agent orders a test absent from the record. In the mercury case, the simulated result varies widely across runs. In the CT case, the simulator responds inconsistently to the same request across runs. Both are most pronounced when the record is sparse relative to what a thorough agent might order. The criticality recall reward partially mitigates the downstream effect, since facts are extracted from the patient record and simulated test results do not correspond to any entry in the fact list, yielding no reward signal and limiting the extent to which the agent is reinforced for pursuing absent tests.

We explored an alternative simulator configuration that strictly declines any test not documented in the record, but found that the resulting observations were too sparse to sustain meaningful exploration during training, as agents received little signal to guide subsequent actions. The current design therefore accepts some stochasticity in exchange for richer exploratory signal. This stochasticity is inherent to the open-ended simulation setting. While simulated results for absent tests vary across runs, the validation below shows that they remain clinically reasonable in practice.

C.3 Simulator Validation

We evaluated simulator output quality on the full test set of 156 cases (75 MIMIC-CDM, 81 Rare Disease). Two physician raters reviewed each step of each trajectory and flagged any step meeting one or more of the following criteria:

- **(1) Contradicts established case information** — introduces facts that directly conflict with what has already been presented (e.g., the HPI states no prior surgeries, but the physical exam result mentions a visible laparotomy scar).
- **(2) Clinically impossible or implausible** — values or findings that could not realistically occur (e.g., sodium of 280 mEq/L; pulse oximetry of 15% in an awake, conversational patient).
- **(3) Wrong result for the ordered test** — the result does not correspond to the test ordered (e.g., a CBC was ordered but a lipid panel was returned; an MRI Brain was ordered but an echocardiogram report is returned).
- **(4) Internally inconsistent** — the result contradicts itself within the same response (e.g., “Chest X-ray shows no pneumothorax” in the findings but “large left pneumothorax” in the impression; or “WBC is within normal limits (38,000/ μL)”).

Table 6: Simulator step validation. A step is flagged if it meets any of the four quality criteria.

Dataset	Cases	Steps	Step-level		Flag type (steps)			
			Flagged	Rate	(1)	(2)	(3)	(4)
MIMIC-CDM	75	155	4	2.6%	0	0	0	4
Rare Disease	81	239	3	1.3%	3	0	0	0
All	156	394	7	1.8%	3	0	0	4

Each case consists of multiple simulator steps (one per diagnostic action taken by the agent), so the step count exceeds the case count. MIMIC-CDM cases average 2.1 steps per case (155 steps across 75 cases) and rare disease cases average 3.0 steps per case (239 steps across 81 cases). Flag types (1)–(4) correspond to the four quality criteria listed above.

Of 394 total steps, 7 were flagged (1.8%). Critically, the simulator never produced clinically impossible values or wrong test results, the failure modes that would corrupt training signal. The 7 flagged steps were minor consistency errors (criteria 1 and 4). These results support the validity of the simulator as a training environment.

D Fact Extraction and Criticality Scoring

D.1 Atomic Fact Extraction

Facts are extracted offline before any training and stored directly in the dataset, avoiding LLM calls at training time. The two datasets differ substantially in their data structure and information density, which motivates different extraction strategies. Rare disease cases are short narrative case reports with a mean of 9.7 facts per case (median 8.5, std 5.5, range 1–35), making a single prompt sufficient to process the full context. MIMIC-CDM cases are structured records with a mean of 131.1 facts per case (median 122.5, std 46.6, range 41–401). Table 7 summarizes the fact statistics for both datasets.

Table 7: Fact extraction statistics for both datasets. The difference in mean facts per case reflects the difference in source data: rare disease cases are short case reports, while MIMIC-CDM cases are full records with structured laboratory, radiology, and examination data.

Metric	Rare Disease (268 cases)	MIMIC-CDM (250 cases)
Total facts	2,609	32,772
Mean / case	9.7	131.1
Median / case	8.5	122.5
Std	5.5	46.6
Min / Max	1 / 35	41 / 401

Rare Disease. Each case is processed with a single LLM call that applies a general-purpose extraction prompt to the full context field. The prompt instructs the model to extract objective findings from diagnostic testing (imaging, endoscopy, pathology, laboratory results) and physical examination, while excluding patient initial presentation, demographic details, and interpretations of test results. Multi-fact sentences are split into atomic items, where each item is a single observation stated without inference. The extraction model is Claude Sonnet 4.5 (prompt in Appendix L.2).

MIMIC-CDM. The MIMIC-CDM context is structured as a JSON record with distinct clinical modalities, so extraction uses three separate strategies. The extraction model is Claude Sonnet 4.5 for both physical examination and radiology (prompts in Appendix L.2).

Physical examination. A dedicated physical examination prompt is applied to the examination field of the context. Included findings are vital signs and explicit exam findings from auscultation, palpation, and inspection, along with negative findings when explicitly stated. Demographics, history, symptoms, and interpretive conclusions are excluded.

652 *Radiology.* Each radiology report in the case is sent to the LLM independently, one call per report.
653 Isolating reports prevents findings from one imaging study from suppressing or contaminating those
654 of another when multiple studies are present. The radiology prompt extracts both positive and
655 negative findings and quantitative measurements, while excluding interpretive conclusions.

656 *Laboratory tests and microbiology.* Lab and microbiology values are stored as structured key-value
657 dictionaries in the MIMIC context and are extracted deterministically without an LLM call.

658 The three modality fact lists are concatenated in order (physical examination, radiology, laboratory
659 and microbiology) to form the final fact list for each case.

660 D.2 Fact Criticality Scoring

661 For both datasets, we use GPT-5.1 to score each fact on a 0–3 integer scale, scoring for how critical
662 the fact is in confirming the gold diagnosis rather than for what diagnosis it suggests. The prompts
663 for both datasets are provided in Appendix L.4 and Appendix L.3.

664 **Rare disease scoring.** Rare disease cases use a simpler prompt with no injected diagnostic criteria.
665 At an average of 10 facts per case drawn from curated case report workups, the scoring task is smaller
666 and less ambiguous. Each case is a distinct rare condition with its own unique presentation, so
667 disease-specific criteria would not generalize across cases. The scoring scale uses general definitions
668 without disease-specific examples.

669 **MIMIC-CDM scoring.** MIMIC cases require a more carefully designed prompt because of scale
670 and ambiguity. At an average of 131 facts per case, most of which are routine lab values and negative
671 imaging findings, the model needs specific clinical anchors to distinguish signal from noise. The
672 three diseases also share overlapping acute abdominal presentations, making it easy for the model to
673 apply the wrong scoring logic.

674 To address this, the MIMIC prompt injects a disease-specific diagnostic criteria block selected by
675 the case’s disease label (cholecystitis, diverticulitis, or pancreatitis), drawn from established clinical
676 guidelines. The gold diagnosis takes priority over the injected criteria, which serve as a reference
677 framework rather than a strict checklist. The full criteria blocks are provided in Appendix L.4. Table 8
678 summarizes the diagnostic structure for each disease category.

Table 8: Comparison of diagnostic structures across disease categories.

Disease	Diagnostic Rule	Clinical Signs	Lab Evidence	Imaging Role
Cholecystitis	Local + systemic signs; definitive diagnosis requires imaging (Tokyo Guidelines)	Murphy sign, right upper quadrant pain/tenderness	Fever, elevated WBC, elevated CRP	Required for definitive diagnosis
Diverticulitis	Clinical suspicion with imaging confirmation	Left lower quadrant pain, tenderness, fever, history	Elevated CRP (supporting)	CT typically used to confirm diagnosis and assess complications
Pancreatitis	Diagnosis requires ≥ 2 of: abdominal pain, enzyme elevation, or imaging findings	Upper abdominal pain	Amylase or lipase $\geq 3 \times$ normal	Optional confirmation if diagnosis uncertain

679 The MIMIC prompt also provides explicit guidance on negative and absent findings, which the
680 radiology extraction pipeline preserves in large volume. Absent findings (e.g., “no free air”, “normal
681 appendix”) default to score 0, but score 1 when the absence specifically confirms the variant stated in
682 the gold diagnosis. For example, “no free air” scores 1 for uncomplicated diverticulitis (confirms the
683 uncomplicated nature).

684 **Score distributions.** Figure 7 shows the observed score distributions. The 88.1% zero rate in
685 MIMIC-CDM reflects that routine laboratory panels and negative CT findings are correctly identified
686 as non-confirmatory. The more balanced rare disease distribution (31% score 0) reflects that case
687 report contexts are already curated to contain the clinically relevant workup.

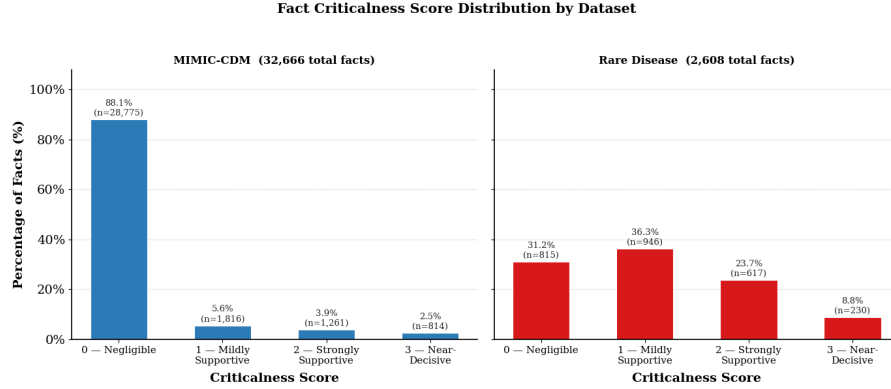


Figure 7: Criticalness score distributions for both datasets. MIMIC-CDM is heavily zero-skewed (88% score 0) because most extracted facts are routine lab values; rare disease cases have a more balanced profile (31% score 0) because case reports contain only the clinically relevant workup.

D.3 Criticality Scorer Validation

We validated the automated criticality scorer against two physician raters on 156 cases (75 MIMIC-CDM, 81 Rare Disease). For each fact in each case, the physicians indicated whether they agreed with the assigned score. Disagreement rates were 1.0% of facts on MIMIC-CDM and 5.7% on Rare Disease (1.3% overall). The higher rate on Rare Disease partly reflects sample size, making the rate more sensitive to individual disagreements. These results indicate that the automated scorer is well-validated against physician judgment across both datasets.

E Evaluation Protocol

E.1 LLM Judges

Two separate LLM graders are used, one for diagnostic accuracy and one for criticality recall.

Diagnosis grader. Final diagnoses are evaluated using Claude Sonnet 4 as the grading model. Given the LLM-generated diagnosis and the ground truth discharge diagnosis, the grader reasons over clinical equivalence and outputs a binary score (1.0 correct, 0.0 incorrect). For MIMIC-CDM, discharge procedures are also provided as supporting evidence for severity and treatment intent.

Fact matching grader. Criticality recall requires determining which atomic facts from the scored fact set were discovered during a trajectory. This matching is performed using google/gemini-3-flash-preview. Given the full interaction history and the list of atomized facts, the grader identifies which facts were revealed during the trajectory and returns their indices, used to compute criticality recall as described in Section 4.

The full prompts for both graders are provided in Appendix L.5.

E.2 Diagnosis Grader Prompt Design

The two datasets require different levels of grader design effort.

Rare Disease grader. Rare disease labels are highly specific and largely unambiguous. Conditions such as leiomyosarcoma or polycystic liver disease admit no clinically meaningful near-miss. The differential diagnosis space is so large that any wrong answer is a complete miss. Grading therefore reduces to a straightforward identity check of whether the model identified the correct rare condition.

That said, rare diseases can carry multiple valid names in the clinical literature, and the grader must still recognize terminological equivalence. For example, *emphysematous gastritis* and *gastric cystic pneumatosis* describe the same condition (gas within the gastric wall), and a prediction of one should be accepted when the ground truth uses the other. The grader prompt is designed to handle this

718 through semantic reasoning rather than string matching, without requiring the elaborate clinical
719 equivalence rules needed for MIMIC-CDM.

720 **MIMIC-CDM grader.** MIMIC-CDM presents a harder grading problem despite its terse labels.
721 The three conditions (cholecystitis, diverticulitis, pancreatitis) are clinically related acute abdominal
722 presentations that can overlap, and the ground truth labels are short phrases that hide substantial
723 clinical variation. Terms such as *acute cholecystitis*, *calculous cholecystitis*, and *acute calculous*
724 *cholecystitis* all refer to the same condition, while *biliary colic* and *gangrenous cholecystitis* do
725 not, yet all are superficially similar. The grader must reason about lexical variation, specificity,
726 severity, and clinical equivalence before it can answer the basic yes/no question. This is why the
727 MIMIC-CDM grader was designed through physician consultation and validated on an explicit test
728 suite, as described below.

729 The diagnosis grader prompt was developed through consultation with a physician to resolve ambigu-
730 ous grading cases that arise from the open-ended label space. The core grading question is whether a
731 clinician acting on the predicted diagnosis would arrive at the same treatment decisions as one acting
732 on the ground truth. The following principles were established:

733 • **Specificity differences are only meaningful if they change treatment.** A prediction
734 more or less specific than the gold is not automatically incorrect. *Sigmoid diverticulitis*
735 and *acute diverticulitis of the sigmoid colon* are both acceptable for a gold of *acute di-*
736 *verticulitis* since management is identical, but predicting *acute cholecystitis* for a gold
737 of *gangrenous cholecystitis* is incorrect because gangrenous cholecystitis requires urgent
738 surgical intervention.

739 • **Related diseases are not equivalent.** Conditions that share a mechanism or anatomi-
740 cal region but have distinct management pathways are graded incorrect. *Biliary colic*
741 and *symptomatic cholelithiasis* are not acceptable for a gold of *acute cholecystitis*, nor is
742 *choledocholithiasis with biliary obstruction*, which involves the bile duct rather than the
743 gallbladder.

744 • **Extra diagnoses are acceptable unless they redirect care.** If a prediction correctly
745 identifies the primary diagnosis and appends additional conditions, it is graded correct
746 provided the added diagnosis does not redirect care to a fundamentally different pathway.
747 Adding a closely related complication within the same organ system is not sufficient grounds
748 to fail a prediction.

749 • **For multi-diagnosis gold labels, the primary diagnosis must be present.** When the
750 gold contains multiple conditions, the prediction must cover the primary (first-listed) one.
751 Predicting only a secondary condition while missing the primary is incorrect. Procedures
752 are used as soft evidence for severity and treatment intent but do not override gold label
753 ordering to reclassify which diagnosis is primary.

754 Table 9 illustrates these principles with representative cases.

Table 9: Selected grading cases across five categories (17/17 passed). Grade 1.0 = correct, 0.0 = incorrect. Etiology matters only when it changes the treatment pathway.

Gold	Prediction	Grade	Reasoning
<i>Specificity & Severity</i>			
Gangrenous cholecystitis	Acute cholecystitis	0.0	Misses severity; gangrenous carries high mortality and requires urgent surgery
Perforated diverticulitis	Acute sigmoid diverticulitis with mild inflammation	0.0	Severity downgrade; perforation confirmed by laparotomy
Diverticulitis	Perforated diverticulitis with pneumoperitoneum	0.0	Severity upgrade; implies emergency surgery when gold warrants antibiotics only
Diverticulitis	[verbose paragraph concluding:] acute uncomplicated sigmoid diverticulitis	1.0	Same diagnosis regardless of narrative framing; management unchanged
<i>Related Diseases</i>			
Acute cholecystitis	Symptomatic cholelithiasis (biliary colic)	0.0	Distinct disease; biliary colic is managed expectantly, not surgically
<i>Extra Diagnoses</i>			
Acute cholecystitis	Acute calculous cholecystitis with mild gallstone pancreatitis	1.0	Extra complication in same biliary domain; does not redirect care
Acute alcoholic pancreatitis	Acute alcoholic pancreatitis with suspected pancreatic adenocarcinoma	0.0	Added cancer diagnosis redirects to fundamentally different clinical pathway
<i>Multi-Diagnosis</i>			
Acute cholecystitis + cholelithiasis	Acute cholecystitis	1.0	Primary diagnosis covered; cholelithiasis is implied by calculous cholecystitis and does not change management
<i>Etiology</i>			
Gallstone pancreatitis	Acute pancreatitis	0.0	Gallstone etiology omitted; triggers cholecystectomy, changing management
EtOH pancreatitis	Acute pancreatitis	1.0	EtOH etiology omitted; acute management identical (bowel rest, IV fluids, NPO)

E.3 Grader Validation

We validated the diagnosis grader against two physician raters on a sample of 143 cases (71 MIMIC-CDM, 72 Rare Disease). Physician availability constrained the sample size. For each case, the physicians reviewed the LLM grader’s binary correct/incorrect judgment and indicated whether they agreed with it.

Table 10: Diagnosis grader validation against physician rater.

Dataset	N	Agree	Disagree	Agreement %
MIMIC-CDM	71	64	7	90.1
Rare Disease	72	67	5	93.1
All	143	131	12	91.6

760 Agreement is 91.6% overall (90.1% on MIMIC-CDM, 93.1% on Rare Disease), indicating that the
761 grader’s binary judgments are consistent with physician assessment across both datasets.

762 F Baseline Implementation Details

763 F.1 Closed-Weight Model Evaluation

764 **Diagnostic loop.** Each evaluation episode runs the policy LLM against the clinical simulator (Claude
765 Sonnet 4, held fixed across all experiments) for at most 7 turns. The policy never has direct access to
766 the patient record. It observes only the action–result history that accumulates turn by turn. At each
767 turn it must output exactly one line, either “Diagnosis: <diagnosis>” to commit to a final answer
768 and terminate the episode, or “Next Action: <action>” to request the next clinical test. The action
769 selection prompt is provided in Appendix L.6.

770 **Post-hoc differential.** A structured differential diagnosis is generated *once* at loop termination by
771 prompting the policy on the completed history. If the agent has not committed to a diagnosis after
772 6 actions, the top entry of the differential is taken as the fallback final diagnosis. The differential
773 generation prompt is provided in Appendix L.6.

774 **Models evaluated.** The policy is instantiated with each of the following models accessed via
775 OpenRouter: GPT-5, GPT-5.1, GPT-5.4, Claude Sonnet 4, Claude Sonnet 4.5, Claude Opus 4.7,
776 Gemini 2.5 Pro, and Gemini 3.1 Pro. The simulator is always Claude Sonnet 4 regardless of which
777 model serves as the policy.

778 F.2 Oracle Context

779 The oracle context baseline evaluates GPT-5.1 in a one-shot setting where the full patient record is
780 provided directly, with no sequential workup and no simulator. It establishes an upper bound on
781 diagnostic accuracy. The model sees the complete information state a clinician would have access
782 to, so any gap relative to the simulation pipeline reflects the cost of having to acquire evidence
783 interactively.

784 The full patient context field is passed directly to the differential diagnosis prompt (the same Top-
785 K framing used in the action simulation, see Appendix L.6). The model produces a structured
786 differential, and the top entry is taken as the final diagnosis. No simulator is invoked and steps taken
787 is always 0.

788 G Training Details

789 All training is run through Tinker [11], the managed training API from Thinking Machines, which
790 handles GPU allocation and orchestration. We therefore report training configuration (LoRA rank,
791 GRPO settings, reward parameters, optimizer) rather than raw hardware specifications.

792 G.1 Models

793 We fine-tune two base models, Kimi-K2.5 and Qwen3-32B, both with LoRA rank 32. Both are
794 trained with GRPO on the same reward function. Dataset-specific and model-specific differences are
795 noted below where they arise.

796 G.2 GRPO Configuration

797 Each GRPO update step samples 6 patient cases and draws 8 rollouts per case (48 effective rollouts
798 per step). Both models are trained for 2 epochs on MIMIC-CDM and 1 epoch on the rare disease
799 dataset. The agent and simulator prompts used during rollouts are provided in Appendix L.7.

800 G.3 Reward Parameters

801 Table 11 lists the concrete parameter values used in the reward function $R(\tau) = (\alpha \cdot CR(\tau) +$
 802 $\beta) \text{acc}(\hat{d}, d^*) - \lambda T/T_{\max}$ across all configurations. Diagnosis correctness is graded by Claude
 803 Sonnet 4 (binary), and criticality recall is computed by google/gemini-3-flash-preview.

Table 11: Reward function parameter values.

Parameter	Description	Value
α	Criticality recall multiplier on correct diagnosis	1.0
β	Base reward for correct diagnosis	0.5
λ	Step penalty weight	0.05
Malformed penalty	Subtracted for invalid output format	0.3

804 G.4 Optimizer and Generation

805 Both models use AdamW. Kimi-K2.5 uses a constant learning rate of 5×10^{-6} , and Qwen3-32B
 806 uses a constant learning rate of 1×10^{-5} . During training, rollouts are sampled at temperature 0.95,
 807 and validation uses greedy decoding. Maximum new tokens per turn is 6,144 for both Kimi-K2.5 and
 808 Qwen3-32B. Episodes terminate when the agent issues a diagnosis, produces malformed output, or
 809 exceeds the maximum of 7 turns.

810 G.5 Evaluation

811 At test time, both RL-trained models are evaluated under greedy decoding (temperature 0.0) with
 812 the same simulator, episode structure, and turn limit as during training. The untrained base model
 813 (reported as *Base* in the results) is evaluated on the identical setting with the same prompts, simulator,
 814 and 7-turn maximum. This ensures that any difference in performance between the base and
 815 RL-trained models is attributable solely to the policy update and not to differences in inference
 816 configuration.

817 H Proof of Proposition 1

818 **Proposition** (Proposition 1, restated). Assume $\sum_{f \in F} c(f) > 0$. Let each fact $f \in F$ have latent
 819 relevance $r_f \geq 0$, and suppose the scorer satisfies

$$c(f) = \kappa r_f + \varepsilon_f, \quad \kappa > 0, \quad |\varepsilon_f| \leq \sigma,$$

820 where $\sigma \geq 0$ is the worst-case per-fact scoring error. For two trajectories $\tau^{(1)}, \tau^{(2)}$ from the same
 821 patient case (sharing fact set F) with discovered sets $D_i = D(\tau^{(i)})$, define $\nu(D) = \sum_{f \in D} r_f$. If

$$\nu(D_1) - \nu(D_2) > \frac{\sigma}{\kappa}(|D_1| + |D_2|),$$

822 then $CR(\tau^{(1)}) > CR(\tau^{(2)})$. Furthermore, if $\alpha > 0$, both trajectories are correct, and $CR(\tau^{(1)}) -$
 823 $CR(\tau^{(2)}) > \frac{\lambda}{\alpha} \cdot \frac{T_1 - T_2}{T_{\max}}$, where T_i is the number of actions in $\tau^{(i)}$, then $R(\tau^{(1)}) > R(\tau^{(2)})$.

824 *Proof. Part 1:* $CR(\tau^{(1)}) > CR(\tau^{(2)})$. Expanding the weighted sums using $c(f) = \kappa r_f + \varepsilon_f$:

$$\sum_{f \in D_1} c(f) - \sum_{f \in D_2} c(f) = \kappa(\nu(D_1) - \nu(D_2)) + \sum_{f \in D_1} \varepsilon_f - \sum_{f \in D_2} \varepsilon_f.$$

825 Since $|\varepsilon_f| \leq \sigma$, the error terms satisfy $\sum_{f \in D_1} \varepsilon_f \geq -\sigma|D_1|$ and $-\sum_{f \in D_2} \varepsilon_f \geq -\sigma|D_2|$, so

$$\sum_{f \in D_1} c(f) - \sum_{f \in D_2} c(f) \geq \kappa(\nu(D_1) - \nu(D_2)) - \sigma(|D_1| + |D_2|).$$

826 Since $\kappa > 0$, the hypothesis gives $\kappa(\nu(D_1) - \nu(D_2)) > \sigma(|D_1| + |D_2|)$, so the right-hand side
 827 is strictly positive. Since $CR(\tau^{(i)}) = \frac{\sum_{f \in D_i} c(f)}{\sum_{f \in F} c(f)}$ and the denominator is shared and positive by
 828 assumption, it follows that $CR(\tau^{(1)}) > CR(\tau^{(2)})$.

829 **Part 2:** $R(\tau^{(1)}) > R(\tau^{(2)})$. For correct trajectories ($\text{acc}(\hat{d}_i, d^*) = 1$) with action counts T_1 and T_2 ,
830 expanding the reward (noting that β cancels) gives

$$R(\tau^{(1)}) - R(\tau^{(2)}) = \alpha(CR(\tau^{(1)}) - CR(\tau^{(2)})) - \lambda \frac{T_1 - T_2}{T_{\max}}.$$

831 Since $\alpha > 0$ and by hypothesis $CR(\tau^{(1)}) - CR(\tau^{(2)}) > \frac{\lambda}{\alpha} \cdot \frac{T_1 - T_2}{T_{\max}}$, it follows that $R(\tau^{(1)}) -$
832 $R(\tau^{(2)}) > 0$. In particular, when Part 1 holds the CR gap is positive, so Part 2 requires only that this
833 gap not be overwhelmed by the step-penalty difference. $\square$

834 *Remark.* Two aspects of the proposition deserve comment. First, on the step-penalty term: with
835 $\alpha = 1.0$, $\lambda = 0.05$, and $T_{\max} = 7$, the maximum value of $\frac{\lambda}{\alpha} \cdot \frac{T_1 - T_2}{T_{\max}}$ is 0.05, so the reward ordering
836 is overturned only when two correct trajectories differ in criticality recall by less than five percentage
837 points. Second, on the scoring noise: since $c(f) \in \{0, 1, 2, 3\}$, the worst-case per-fact error is $\sigma \leq 3$
838 (a scorer that assigns 0 to every hallmark fact, or 3 to every irrelevant one), and the Part 1 sufficient
839 condition becomes harder to satisfy as σ grows. The bound is vacuous at $\sigma = 3$ and most informative
840 when σ is well below this maximum, for instance $\sigma = 1$ (the scorer is at most one level off on any
841 fact). Empirical validation of the scorer is reported in Appendix D.2.

842 I Failure Case Study

843 Case: Gastric metastasis from uveal melanoma

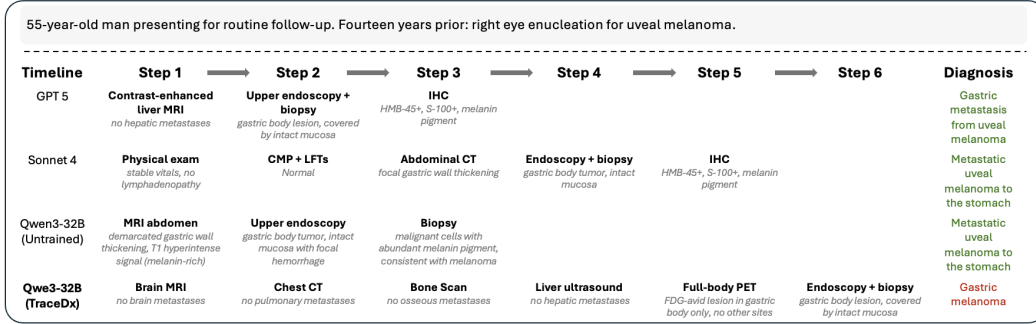


Figure 8: Failure case: diagnostic trajectories for gastric metastasis from uveal melanoma. The grey box (top) shows the initial patient presentation; trajectories below show actions and results. Baseline models correctly anchor on the patient’s prior uveal melanoma and reach the correct diagnosis. TraceDx initiates a broad staging workup, finds no distant metastases, and incorrectly concludes primary gastric melanoma, failing to integrate the patient’s history as a prior constraint.

844 A 55-year-old man with a history of uveal melanoma treated 14 years prior presented for routine
845 follow-up. Imaging revealed a T1-hyperintense gastric wall lesion, and endoscopy with biopsy
846 confirmed melanoma. Uveal melanoma is known to metastasize years or even decades after treatment,
847 most commonly to the liver but also to the gastrointestinal tract. The central diagnostic question was
848 therefore whether this lesion represented delayed metastatic disease rather than a new primary tumor.

849 Baseline models correctly anchored on the patient’s history and interpreted the gastric lesion as
850 metastatic uveal melanoma. They used imaging and biopsy to confirm the presence of melanoma but
851 consistently treated the prior malignancy as the most likely source, leading to the correct diagnosis.

852 In contrast, the trained model initiated a comprehensive staging workup, including brain MRI, chest
853 CT, bone scan, liver imaging, and PET. These tests showed no evidence of disease outside the stomach.
854 Based on this, the model concluded that the gastric lesion represented a primary melanoma. However,
855 primary gastric melanoma is extraordinarily rare and is a diagnosis of exclusion that requires ruling
856 out all known primary sites, including prior uveal melanoma.

857 This failure reflects a breakdown in information integration rather than knowledge or execution. The
858 model correctly identified melanoma on biopsy and performed an appropriate staging protocol, but

859 did not incorporate the patient's prior uveal melanoma into its reasoning. Instead, it treated the case
860 as a new malignancy and used the absence of widespread disease to infer a primary origin.

861 Notably, the error was driven by otherwise correct actions. The staging workup was clinically
862 appropriate, but by prioritizing it over the patient's history, the model allowed the absence of
863 metastases to override a far stronger prior. In this case, more comprehensive testing led to a less
864 accurate conclusion. The model did not fail to gather information. It failed to use the most important
865 information it already had.

866 **Case: Acute gallstone pancreatitis**

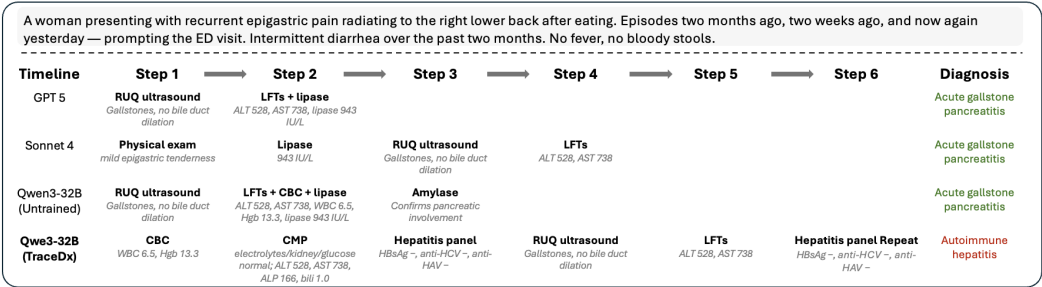


Figure 9: Failure case: diagnostic trajectories for acute gallstone pancreatitis. The grey box (top) shows the initial patient presentation; trajectories below show actions and results. Baseline models rapidly identify gallstones and measure lipase, correctly diagnosing gallstone pancreatitis. In contrast, the trained model follows a hepatology-focused workup, repeatedly evaluating for hepatitis and never measuring lipase, leading to an incorrect diagnosis of autoimmune hepatitis.

867 A woman presented with recurrent post-prandial epigastric pain radiating to the right lower back,
868 a pattern suggestive of biliary disease. Imaging demonstrated gallstones, and laboratory results
869 showed markedly elevated transaminases (ALT 528 IU/L, AST 738 IU/L). The central diagnostic
870 question was whether these abnormalities reflected primary liver disease or secondary effects of
871 biliary obstruction involving the pancreas.

872 Baseline models followed a short and targeted reasoning chain. After identifying gallstones, they
873 measured serum lipase, which was markedly elevated (943 IU/L, nearly 16× the upper limit of
874 normal), confirming pancreatic involvement and leading directly to the diagnosis of acute gallstone
875 pancreatitis. In each case, the models prioritized the pancreatic hypothesis early and selected the test
876 that could resolve it.

877 The trained model instead pursued a hepatology-focused workup. Markedly elevated transaminases
878 alongside gallstones are a legitimate reason to investigate primary liver disease, and ruling out viral
879 hepatitis is a clinically reasonable step. The model did this correctly. But it never closed the loop on
880 pancreatic involvement, and without a lipase measurement, the elevated liver enzymes became the
881 dominant signal, leading to a conclusion of autoimmune hepatitis.

882 The failure is therefore one of completeness rather than reasoning. The hepatology workup was not
883 wrong, but it was pursued to the exclusion of a simpler test that would have resolved the case. In
884 gallstone pancreatitis, elevated transaminases are a secondary effect of transient biliary obstruction,
885 and measuring lipase is the step that distinguishes this from primary liver disease. The trained model
886 never took that step.

887 **J Physician Baseline**

888 To contextualize model performance, we evaluated physicians playing the same simulator game as
889 the models, subject to the same turn limit and interface. Due to physician time constraints, evaluation
890 was conducted on a smaller subset of the held-out test set. Physicians interacted with the clinical
891 simulator by issuing free-text diagnostic actions and committing to a final diagnosis, with no access
892 to the patient record.

Table 12: Physician and model accuracy on randomly sampled subsets of each test set under the same simulator game. Rare Disease: 60 cases, 2 physicians (30 each). MIMIC-CDM: 30 cases, 1 physician.

Model	Rare Disease	MIMIC-CDM
Physician	31.7%	76.7%
Opus 4.7	41.7%	67.8%
GPT-5.4	33.3%	64.4%
Kimi-K2.5 TraceDx	37.8%	65.6%
Qwen3-32B TraceDx	33.3%	62.2%

Rare Disease. Two physicians each evaluated 30 cases randomly sampled from the Rare Disease test set (60 cases total). Physician accuracy was 31.7%, below all models in the comparison. This reflects the intrinsic difficulty of the rare disease task, as the conditions are uncommon enough that even experienced clinicians may not recognize the hallmark findings without the breadth of training data available to large language models.

MIMIC-CDM. One physician evaluated 30 cases randomly sampled from the MIMIC-CDM test set (10 per disease category). Physician accuracy was 76.7%, exceeding all models. On familiar acute abdominal presentations, clinicians apply well-practiced diagnostic heuristics that current models have not yet matched in the interactive setting.

The two datasets present opposite orderings, with physicians trailing models on rare diseases but leading on common abdominal conditions. On Rare Disease, both TraceDx variants match or exceed physician performance, with Kimi-K2.5 TraceDx leading by 6.1 percentage points. On MIMIC-CDM, TraceDx trails the physician but remains competitive with frontier models, with Kimi-K2.5 TraceDx surpassing GPT-5.4.

K Impact Statement

The proposed framework, TraceDx, advances sequential clinical diagnosis by training agents to recover the evidence that drives diagnosis. By decomposing patient records into atomic clinical findings and rewarding agents for retrieving high-criticality evidence, TraceDx provides a fine-grained training signal that aligns with how clinicians weigh evidence in practice. This contribution is particularly impactful in domains where decisions hinge on identifying decisive findings within heterogeneous, noisy patient data, and demonstrates that open-weight models can match or exceed multiple frontier baselines when supervised by an evidence-aware reward.

The primary benefit of this work is methodological. TraceDx shows that fact-level diagnostic criticality is a tractable and effective training signal for sequential clinical reasoning, enabling open-weight diagnostic agents that match or exceed multiple frontier baselines on both common and rare conditions. The fact-atomization and criticality-scoring pipeline also extends naturally to other expert domains where decisions hinge on evidence quality, such as legal reasoning, scientific investigation, or technical troubleshooting.

As with any machine learning method applied to healthcare, careful evaluation precedes any deployment. TraceDx is developed and evaluated within a clinical simulator grounded in real patient records and published case reports, providing a reproducible and validated training environment (Appendix C.3). Performance in this setting reflects diagnostic reasoning under faithful evidence, and transfer to live clinical practice would benefit from prospective validation, regulatory review, and clinician oversight following the standard pathway for medical AI. The training corpora also span a specific patient population (MIMIC-IV draws from a single US tertiary care system) and a specific disease area (abdominal pathologies in MIMIC-CDM, gastrointestinal and hepatobiliary conditions in the rare disease set), and broader clinical generalization will benefit from continued evaluation across additional sites, populations, and disease categories. We position TraceDx as a research artifact intended to advance the methodology of clinical diagnostic AI.

932 L Prompt Templates

933 L.1 Simulator Prompts

Simulator System Prompt for Rare Disease Dataset

We are interested in simulating a medical diagnosis pipeline.

Specifically, I am going to give you prior steps that have been taken towards diagnosis, and also the most recent action that the doctor has taken, which might be a physical exam or medical imaging.

You will also be given access to the ground truth patient file.

This data is the result of the true medical diagnosis pipeline that occurred at a real hospital.

The doctor should not have access to this data.

You only have it to help inform your simulation.

We are trying to train new doctors. And part of this is simulating results of medical exams and tests.

Your job is to look at the ground truth patient file, along with the doctor's past actions and current action, and output what information would likely be revealed by the doctor's current action.

Please ground your simulation in the patient's ground truth data, as that document is a ground truth record of what actually happened.

Do not mention this ground truth data in your simulation.

The doctor does not have access to this data and does not know it exists. It is a secret.

Here is an example of a good simulation:

<begin simulation example>

{simulation_example}

<end simulation example>

Here is the ground truth patient file for simulation assistance:

<begin ground truth patient file for simulation assistance>

{full_patient_file}

<end ground truth patient file for simulation assistance>

Please only use the ground truth patient file to simulate the next step.

Note that we are trying to test the doctor's ability to treat, so you don't want to leak anything from this ground truth file.

This file was only collected at the very end of a medical case.

It is not the patient's current file or medical history.

It is a tool to help you with simulation.

If you leak information from this file, you will ruin the simulation

and doctors will not learn anything from this exercise.

Think about what information would be revealed by the doctor's action, and don't disclose any more than that information.

Here is the list of prior actions the doctor has taken and their results:

<begin prior actions>

{action_history}

<end prior actions>

Here is the most recent action the doctor has taken:

<begin most recent action>

{action}

<end most recent action>

Please use the patient's full medical history, along with the doctor's action, to simulate what information will be revealed by the doctor's action.

Please output only the results of your simulation. Keep things concise.

Simulation Results:

934

Simulator System Prompt for MIMIC-CDM

We are interested in simulating a medical diagnosis pipeline.

Specifically, I am going to give you prior steps that have been taken towards diagnosis, and also the most recent action that the doctor has taken, which might be a physical exam or medical imaging.

You will also be given access to the ground truth patient file.

This data is the result of the true medical diagnosis pipeline that occurred at a real hospital.

The doctor should not have access to this data.

You only have it to help inform your simulation.

We are trying to train new doctors. And part of this is simulating results of medical exams and tests.

Any underlines or blanks (e.g., "___", "____") in the patient case file indicate intentionally redacted information.

These redactions are normal and reflect either sensitive information or details that would directly reveal the ultimate diagnosis.

Do NOT attempt to infer, redacted information. It is acceptable for the simulation output to include underlines or blanks (e.g., "___", "____") when the corresponding information is redacted.

Do NOT attempt to replace or resolve redacted content.

Your job is to look at the ground truth patient file, along with the doctor's past actions and current action, and output what information would likely be revealed by the doctor's current action.

Please ground your simulation in the patient's ground truth data, as that document is a ground truth record of what actually happened.

Do not mention this ground truth data in your simulation.

The doctor does not have access to this data and does not know it exists. It is a secret.

Here is an example of a good simulation:

<begin simulation example>

{simulation_example}

<end simulation example>

Here is the ground truth patient file for simulation assistance:

<begin ground truth patient file for simulation assistance>

{full_patient_file}

<end ground truth patient file for simulation assistance>

Please only use the ground truth patient file to simulate the next step.

Note that we are trying to test the doctor's ability to treat, so you don't want to leak anything from this ground truth file.

This file was only collected at the very end of a medical case.

It is not the patient's current file or medical history.

It is a tool to help you with simulation.

If you leak information from this file, you will ruin the simulation and doctors will not learn anything from this exercise.

Think about what information would be revealed by the doctor's action, and don't disclose any more than that information.

Here is the list of prior actions the doctor has taken and their results:

<begin prior actions>

{action_history}

<end prior actions>

Here is the most recent action the doctor has taken:

<begin most recent action>

{action}

<end most recent action>

Please use the patient's full medical history, along with the doctor's action, to simulate what information will be revealed by the doctor's action.

Please output only the results of your simulation. Keep things concise.

Simulation Results:

Simulation Example Insert

Prior Steps Taken:

- Patient presented with 3-day history of right lower abdominal pain
- Obtained medical history showing 16-year history of Crohn's disease, currently in remission
- No NSAIDs usage reported
- Previous colonoscopy and MR enterography were unremarkable

Doctor's Action:

Physical examination of the abdomen

Simulation Results:

Examination reveals notable tenderness in the right lower quadrant of the abdomen. No rebound tenderness or guarding is present. The rest of the abdominal examination is unremarkable. No masses are palpated. The tenderness is localized and does not appear to be spreading across the abdomen.

936

L.2 Fact Extraction Prompts

Rare Disease

You are a medical expert. Extract ONLY the atomic clinical facts from the patient case.

OUTPUT FORMAT:

- Respond with a JSON list: ["fact 1", "fact 2", ...]
- No numbering, no labels, no explanations.
- JSON list ONLY. No prose before or after.

INCLUDE:

- Objective findings obtained from diagnostic testing (e.g., imaging, endoscopy, pathology, labs).
- Objective findings from physical examination.
- Results of prior diagnostic evaluations.

EXCLUDE:

- Patient-reported symptoms or subjective complaints.
- Demographic details.
- Any interpretations, summaries, or reasoning not explicitly stated.
- Historical background information that is not an objective test or exam result.
- Any invented content or unstated inference.

RULES:

- Each item must contain exactly ONE atomic fact.
- If a sentence contains multiple facts, split them into separate items.
- Only include facts directly and explicitly stated in the case.

Your output must be a valid JSON array.

CASE:

{case}

938

MIMIC-CDM Physical Exam

You are a medical expert. Extract ONLY atomic clinical facts from the physical examination text below.

OUTPUT:

- A JSON list: ["fact 1", "fact 2", ...]
- JSON list ONLY. No numbering, labels, or explanations.

INCLUDE:

- Explicit physical exam findings (vital signs, auscultation, palpation, inspection findings)
- Negative or absent findings when explicitly stated (e.g., "no rebound tenderness", "no guarding", "no edema")

EXCLUDE:

- Demographics, history, symptoms reported by the patient
- Interpretive conclusions (e.g., "consistent with peritonitis")
- Administrative notes
- Any content not explicitly stated in the text

RULES:

- One atomic fact per item
- Use wording as close to the original as possible
- Do NOT fill in "____", keep the original text as is
- Split multiple findings into separate items
- Do NOT rephrase, normalize, or invent information

939

TEXT:
{case}

940

MIMIC-CDM Radiology

You are a medical expert. Extract ONLY atomic clinical facts from the radiology report text below.

OUTPUT:

- A JSON list: ["fact 1", "fact 2", ...]
- JSON list ONLY. No numbering, labels, or explanations.

INCLUDE:

- Explicit radiology findings (both positive and negative)
- Negative or absent findings when explicitly stated
(e.g., "no free air", "no abscess", "appendix is normal",
"no free gas in the pelvis", "no pericholecystic fluid")
- Observational findings that precede or accompany interpretive conclusions
(e.g., extract "fat stranding surrounding the sigmoid colon" and
"surrounding diverticula" even if the sentence continues with
"suggestive of ____")
- Measurements and quantitative findings
(e.g., "gallbladder wall thickened measuring up to 5 mm",
"common bile duct measures 5 mm")

EXCLUDE:

- Technique descriptions and procedural boilerplate
(e.g., "axial CT images were acquired", "grayscale and Doppler
ultrasound images were acquired", "coronal reformats reviewed")
- Interpretive conclusions and diagnostic inferences
(e.g., "suggestive of diverticulitis", "likely representing X",
"consistent with cholecystitis") -- exclude the conclusion but
keep the underlying observation
- Administrative or specimen handling notes
- Any content not explicitly stated in the text

RULES:

- One atomic fact per item
- Use wording as close to the original as possible
- Do NOT fill in "____", keep the original text as is
- Split multiple findings into separate items
- Do NOT rephrase, normalize, or invent information
- Do NOT invent facts not explicitly stated in the text

TEXT:
{case}
"""

941

942 L.3 Rare Disease Fact Criticality Scoring Prompts

Rare Disease Fact Criticality Scoring

You are a clinical evidence rater.

You will be given:

- 1) The GOLD diagnosis (ground-truth label).
- 2) The patient's INITIAL PRESENTATION (history + symptoms + key context).
- 3) A FACT LIST (individual findings, labs, imaging statements, exam findings).
- 4) NUM_TOTAL_FACTS: The total number of facts in FACT_LIST.

Your job:

Assign a criticalness score (0-3) to EACH fact for how important it is for CONFIRMING the GOLD diagnosis, using ONLY the initial presentation + the content of the fact itself.

IMPORTANT: This is GOLD-CONDITIONED scoring.

- We are NOT asking what diagnosis the fact suggests.
- We are asking: "If the GOLD diagnosis is true, how critical is this fact for establishing/confirming that diagnosis in this case?"

Scoring scale (gold-conditioned):

- 0 = negligible / background:
 - * Non-specific, common, incidental, administrative, or normal findings that do not meaningfully help confirm the GOLD diagnosis.
 - * Facts that are irrelevant or only weakly related.
 - * If a fact would strongly contradict or rule out the GOLD diagnosis, STILL assign 0 (we do not reward contradictions).

943

- 1 = mildly supportive / contextual:
 - * Provides some helpful context consistent with the GOLD diagnosis but is not specific and would not change a clinician's confidence much alone.
 - * Examples: mild leukocytosis, vague abdominal tenderness, general nausea.
- 2 = strongly supportive for confirming the GOLD diagnosis:
 - * A strong, diagnosis-relevant finding that substantially increases confidence in the GOLD diagnosis, but is not by itself nearly sufficient.
 - * Typically a key imaging/lab/exam feature characteristic of the condition (but not a single "smoking gun" that almost forces the diagnosis).
- 3 = near-decisive for confirming the GOLD diagnosis:
 - * A hallmark / defining finding that, together with the initial presentation, would make the GOLD diagnosis highly likely ("almost forced").
 - * Examples (conceptual, not exhaustive): definitive imaging sign of the disease, pathognomonic lab pattern, explicit radiology impression matching the diagnosis, etc.

Guidelines:

- Use the initial presentation to interpret relevance (e.g., timing, location of pain).
- Do NOT reward extra details that are correct but not decision-relevant (e.g., long lists of normal organs on CT).
- If multiple facts repeat the same idea (duplicates), score each independently based on how important it is;
- When uncertain between two scores, choose the LOWER score. Be conservative.
- Keep scoring consistent across the list.
- Make sure to score all facts in the fact list.

Output format (STRICT):

Return ONLY a JSON object (dictionary) where each key is EXACTLY one fact string from FACT LIST and each value is an integer 0-3 (the criticalness score). The JSON object MUST contain exactly NUM_TOTAL_FACTS key-value pairs. Each key MUST be copied VERBATIM from FACT LIST (character-for-character, including punctuation, spacing, and units). Do not add or omit any keys. Use the fact strings exactly as given.

Example output format :

```
{
  "Fact 0 text here": 0,
  "Fact 1 text here": 1,
  "Fact 2 text here": 3,
  "Fact 3 text here": 0,
  "Fact 4 text here": 2
}
```

Do not include explanations, comments, or any additional text.

GOLD DIAGNOSIS:
{gold_diagnosis}

INITIAL PATIENT PRESENTATION:
{initial_presentation}

FACT LIST:
{facts_block}

NUM_TOTAL_FACTS:
{num_total_facts}

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945 L.4 MIMIC-CDM Fact Criticality Scoring Prompts

MIMIC-CDM Fact Criticality Scoring

You are a clinical evidence rater.

You will be given:

- 1) The GOLD DIAGNOSIS (ground-truth label).
- 2) The patient's INITIAL PRESENTATION (history + symptoms + key context).
- 3) DIAGNOSTIC CRITERIA (general diagnostic framework for the relevant disease categories).
- 4) A FACT LIST (individual findings, labs, imaging statements, exam findings).
- 5) NUM_TOTAL_FACTS: The total number of facts in FACT_LIST.

Your job:

Assign a criticalness score (0-3) to EACH fact for how important it is for CONFIRMING the GOLD DIAGNOSIS, using the initial presentation and the fact itself.

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IMPORTANT -- PRIORITY RULE:

- The GOLD DIAGNOSIS is the authoritative ground truth and takes precedence.
- Score facts based on how important they are for confirming what is explicitly stated in the GOLD DIAGNOSIS.
- The DIAGNOSTIC CRITERIA are provided as a general reference framework only.
- Do NOT restrict scoring only to items explicitly listed in the criteria.
- If a fact is clearly critical for confirming any component of the GOLD DIAGNOSIS (including Primary or Secondary diagnoses), it must be scored appropriately even if it is not explicitly emphasized in the criteria.

This is GOLD-CONDITIONED scoring:

- We are NOT asking what diagnosis the fact suggests.
- We are asking: "If the GOLD DIAGNOSIS is true, how critical is this fact for establishing or confirming it in this specific case?".

Scoring scale (gold-conditioned):

- 0 = negligible / background
 - * Non-specific, incidental, administrative, or normal findings that do not meaningfully increase confidence in the GOLD DIAGNOSIS.
 - * Findings unrelated or only very weakly related.
 - * Absent or negative findings (e.g., "no free air", "normal appendix") score 0 UNLESS the absence itself is diagnostically meaningful for the specific GOLD DIAGNOSIS (see score=1 below).
 - * If a fact directly contradicts or rules out the GOLD DIAGNOSIS, assign 0 (do not reward contradictions).
- 1 = mildly supportive / contextual
 - * A POSITIVE finding that is relevant and consistent with the GOLD DIAGNOSIS, but non-specific and would not meaningfully change confidence alone.
 - * Negative/absent findings score 1 ONLY when the absence meaningfully confirms the specific variant stated in the GOLD DIAGNOSIS.
Example: "no free air" in UNCOMPLICATED diverticulitis is score=1 because it confirms the uncomplicated nature. "No free air" in PERFORATED diverticulitis is score=0 because it contradicts it.
- 2 = strongly supportive
 - * A positive finding that substantially increases confidence in the GOLD DIAGNOSIS, but is not by itself nearly sufficient.
 - * Typically a key but non-defining feature -- a finding a clinician would specifically look for and weight heavily when working up this diagnosis, even if it appears in other conditions too.
 - * Examples by disease:
 - Pancreatitis: epigastric pain radiating to the back; elevated WBC or CRP; peripancreatic fat stranding without necrosis; gallstones on imaging when etiology is unspecified.
 - Diverticulitis: focal LLQ tenderness at the correct site; elevated WBC or CRP; prior history of diverticulitis.
 - Cholecystitis: RUQ tenderness or guarding; fever with elevated WBC; positive Murphy sign; gallbladder wall thickening without a confirmatory stone.
- 3 = near-decisive / hallmark
 - * A defining or characteristic finding that directly matches a core diagnostic feature of the GOLD DIAGNOSIS.
 - * Together with the initial presentation, this would make the GOLD DIAGNOSIS highly likely or almost forced.
 - * Examples by disease:
 - Pancreatitis: lipase or amylase $\geq 3 \times$ ULN; pancreatic necrosis or edema on CT; gallstones with dilated CBD when GT states gallstone pancreatitis.
 - Diverticulitis: CT showing wall thickening + fat stranding + diverticula at the correct segment; extraluminal air when GT states perforated; pericolic abscess when GT states abscess.
 - Cholecystitis: gallbladder wall thickening with impacted stone; pericholecystic fluid; interrupted rim sign when GT states gangrenous.

Guidelines:

- Use the initial presentation to interpret relevance.
- Use the diagnostic criteria as guidance, not as a strict checklist.
- Ensure that ALL components of the GOLD DIAGNOSIS are considered.
- If multiple diagnoses are listed in GOLD DIAGNOSIS, a fact should receive a high score if it is critical for confirming ANY of them.
- When uncertain between two scores, choose the LOWER score.
- Score all facts.

Output format (STRICT):
 Return ONLY a JSON object (dictionary) where each key is EXACTLY one fact string from FACT LIST and each value is an integer 0-3 (the criticalness score). The JSON object MUST contain exactly NUM_TOTAL_FACTS key-value pairs. Each key MUST be copied VERBATIM from FACT LIST (character-for-character, including punctuation, spacing, and units). Do not add or omit any keys. Use the fact strings exactly as given.

Example output format:

```
{
  "Fact 0 text here": 0,
  "Fact 1 text here": 1,
  "Fact 2 text here": 3,
  "Fact 3 text here": 0,
  "Fact 4 text here": 2
}
```

Do not include explanations, comments, or any additional text.

GOLD DIAGNOSIS:
 {gold_diagnosis}

INITIAL PATIENT PRESENTATION:
 {initial_presentation}

DIAGNOSTIC CRITERIA:
 {diagnostic_criteria}

FACT LIST:
 {facts_block}

NUM_TOTAL_FACTS:
 {num_total_facts}

948

Pancreatitis Diagnostic Criteria

DIAGNOSTIC CRITERIA -- ACUTE PANCREATITIS
 (Revised Atlanta Classification 2012)

Diagnosis requires at least 2 of the following 3 criteria:

1. Characteristic abdominal pain: epigastric, often radiating to the back.
2. Serum lipase OR amylase $\geq 3\times$ the upper limit of normal (approximately ≥ 150 IU/L for both). Lipase and amylase are equally valid for satisfying this criterion and should be scored equivalently. Note: enzyme level magnitude does not predict severity -- it only confirms diagnosis.
3. Characteristic cross-sectional imaging findings: pancreatic edema or enlargement, peripancreatic fat stranding, fluid collections, or pancreatic necrosis (non-enhancing parenchyma on CT).

SEVERITY -- when the GOLD DIAGNOSIS explicitly states a severe or necrotizing variant, the following are defining features: pancreatic necrosis on imaging, organized peripancreatic fluid collections (pseudocyst, walled-off necrosis), infected necrosis (gas in collection or positive culture), and persistent organ failure (respiratory, cardiovascular, or renal).

ETIOLOGY -- when the GOLD DIAGNOSIS explicitly states an etiology, findings that confirm that etiology are hallmark:

- Gallstone pancreatitis: gallstones or biliary sludge, dilated common bile duct, choledocholithiasis, and elevated liver chemistries (bilirubin, ALT, AST, alkaline phosphatase) at presentation.
- Alcoholic/EtOH pancreatitis: a history of significant, long-term alcohol use is the key etiologic fact. Binge or social drinking alone is not sufficient to establish this etiology.
- Post-ERCP pancreatitis: onset of characteristic pain and enzyme elevation within 24 hours of the procedure is the defining feature.
- Hypertriglyceridemia pancreatitis: markedly elevated triglycerides are the defining etiologic finding.

949

Diverticulitis Diagnostic Criteria

DIAGNOSTIC CRITERIA -- ACUTE DIVERTICULITIS
 (WSES 2020 Guidelines and standard clinical practice)

Diagnosis is based on abdominal pain (typically lower-left quadrant), focal tenderness, low-grade fever, and elevated inflammatory markers (WBC, CRP), confirmed by CT showing colonic wall thickening, pericolonic fat stranding, and diverticula at the affected segment.

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ANATOMICAL SPECIFICITY -- this is the most important scoring principle: Score findings at the anatomical site that matches the GOLD DIAGNOSIS. A finding is only hallmark if it is at the correct location for the stated diagnosis. For example:

- Sigmoid diverticulitis: sigmoid and descending colon findings are hallmark; cecal or right-sided findings are not directly confirmatory.
- Cecal diverticulitis: cecal and right colon findings are hallmark; sigmoid findings are not directly confirmatory.
- Small bowel, transverse colon, ampullary, or duodenal diverticulitis: findings at those specific segments are hallmark; standard sigmoid/colonic findings are not.

UNCOMPLICATED DIVERTICULITIS: wall thickening, fat stranding, and diverticula at the correct segment are the defining CT findings. Meaningful negative findings ("no free air," "no abscess") confirm the uncomplicated nature and are mildly supportive (score=1) when the GOLD DIAGNOSIS states uncomplicated or simple diverticulitis.

COMPLICATED DIVERTICULITIS -- when the GOLD DIAGNOSIS explicitly states a complicated variant, the following are its defining features:

- Perforated diverticulitis: pneumoperitoneum or any extraluminal air, including tiny locules adjacent to the colon or tracking into surrounding structures. Even a small amount of extraluminal air is hallmark for perforation and microperforation variants alike.
- Abscess: a pericolic or intra-abdominal fluid collection (with or without rim enhancement). An inflammatory phlegmon without a drainable collection is also a complication marker.
- Fistula: air in the bladder in the absence of prior instrumentation, or an identifiable fistula tract on imaging, are hallmark for colovesical or colovaginal fistula variants.
- Obstruction: transition point at the affected colonic segment with proximal dilation is hallmark for the obstructing variant.

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Cholecystitis Diagnostic Criteria

DIAGNOSTIC CRITERIA -- ACUTE CHOLECYSTITIS
(Tokyo Guidelines 2018 / TG18, adopted from TG13)

Diagnosis requires at least one local sign of inflammation, one systemic sign of inflammation, and is confirmed by characteristic imaging findings.

Local signs of inflammation:

- Positive Murphy sign (clinical or sonographic)
- Right upper quadrant mass, pain, or tenderness

Systemic signs of inflammation:

- Fever
- Elevated CRP
- Elevated WBC

Characteristic imaging findings:

- Gallbladder wall thickening (>4 mm)
- Pericholecystic fluid
- Gallbladder distension
- Gallstones or biliary sludge, especially a stone impacted in the neck or cystic duct
- Sonographic Murphy sign

COMPLICATED SUBTYPES -- when the GOLD DIAGNOSIS explicitly states a complicated variant, the following are its defining features:

- Gangrenous cholecystitis: irregular or asymmetric wall thickening, interrupted rim sign (loss of normal mural enhancement on contrast CT), intraluminal membranes or flaps, gas in the gallbladder lumen or wall, and pericholecystic abscess. These are hallmark findings for gangrenous variants.
- Emphysematous cholecystitis: gas within the gallbladder wall or lumen on CT is the defining finding.
- Perforated cholecystitis: a visible defect in the gallbladder wall, pericholecystic abscess, or bile peritonitis are hallmark.
- Empyema of the gallbladder: markedly distended gallbladder with purulent contents; positive bile cultures are relevant.
- Acalculous cholecystitis: typical cholecystitis findings in the absence of gallstones. "No gallstones" is a meaningful finding that confirms this variant (score=1) rather than a contradiction.
- Hydrops of the gallbladder: marked distension with an obstructing stone at the neck or cystic duct, without necessarily fulfilling all inflammatory criteria.

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SEVERITY -- when the GOLD DIAGNOSIS states severe cholecystitis or organ dysfunction, findings of organ failure (cardiovascular, respiratory, renal, hepatic, neurological, or hematological) are hallmark for that severity grade. Markedly elevated WBC is a key marker for moderate (Grade II) severity.

ASSOCIATED SECONDARY DIAGNOSES -- when the GOLD DIAGNOSIS lists choledocholithiasis, cholangitis, or biliary obstruction as secondary diagnoses, findings confirming those (CBD dilation, ductal stones, elevated bilirubin and liver enzymes, jaundice) should be scored at their appropriate level for confirming those secondary diagnoses.

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954 L.5 Grader Prompts

MIMIC-CDM Final Diagnosis Grader

We are given an LLM-generated medical diagnosis and a ground truth diagnosis. Optionally, the procedures performed on the patient at discharge may also be provided. We want to judge whether the LLM diagnosis is clinically correct.

This is a nuanced grading task. Two diagnoses can be worded differently and still be clinically equivalent, or they can sound related but be meaningfully distinct. The key question to ask is: would a clinician acting on the LLM's diagnosis arrive at the same treatment decisions as one acting on the ground truth?

=== GRADING PRINCIPLES ===

1. SPECIFICITY DIFFERENCES ARE ONLY MEANINGFUL IF THEY CHANGE TREATMENT.
A prediction that is more or less specific than the gold is not automatically correct or incorrect -- what matters is whether the difference in specificity would lead a clinician to a meaningfully different treatment decision. If the level of detail added or omitted does not change management, the prediction can be considered equivalent. If it does change management (e.g., a missing severity marker like "perforated" or "gangrenous" that triggers a different intervention), the prediction is INCORRECT.
2. RELATED DISEASES ARE NOT EQUIVALENT.
Diseases that share a mechanism or anatomical region but have distinct diagnoses and treatment pathways should be graded INCORRECT. Proximity on a disease spectrum does not make two diagnoses interchangeable.
3. EXTRA DIAGNOSES ARE ACCEPTABLE UNLESS THEY REDIRECT TO A FUNDAMENTALLY DIFFERENT CLINICAL PATHWAY.
If the prediction correctly identifies the primary diagnosis and appends additional conditions, grade as CORRECT. The severity concern applies when the prediction DOWNGRADES the gold (misses a dangerous complication). It does not apply when the prediction ADDS a related complication on top of a correct primary -- that is an enrichment, not a contradiction. Only mark INCORRECT if the added diagnosis would redirect care so substantially that the gold's treatment plan would be abandoned or seriously delayed -- for example, adding a cancer diagnosis to a benign condition. Adding a closely related complication within the same organ system (e.g., mild pancreatitis alongside cholecystitis) is not sufficient grounds to fail the prediction. When in doubt, lean toward CORRECT if the primary is right.
4. FOR MULTI-DIAGNOSIS GOLD LABELS, THE PRIMARY DIAGNOSIS MUST BE PRESENT.
If the gold contains multiple conditions, the prediction must cover the primary one. Missing a secondary condition alone is acceptable. Predicting only a secondary condition while missing the primary is INCORRECT.
IMPORTANT: Do not use procedures to reclassify a secondary gold diagnosis as primary. The ordering of diagnoses in the gold label determines what is primary. Procedures reflect what was done, not necessarily which diagnosis was listed first.
5. USE PROCEDURES AS A SOFT ANCHOR FOR SEVERITY ONLY -- NOT FOR DIAGNOSIS CLASSIFICATION.
Procedures can help confirm whether the gold diagnosis was severe enough to require surgery versus medical management. Use them only for this purpose. Do NOT use procedures to argue that a secondary diagnosis in the gold should have been predicted, or to fail a prediction that correctly identifies the primary. Procedure records may also be incomplete, so their absence does not mean no intervention occurred.

=== WORKED EXAMPLES ===

--- EXAMPLE A: CORRECT -- valid specificity refinement ---

LLM diagnosis:
"Acute sigmoid diverticulitis with small pericolic abscess."

Ground truth:
"Acute diverticulitis"

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Procedures: None documented.

Thought: The gold is general. The LLM adds anatomical specificity (sigmoid) and a complication (small abscess), but both diagnoses point to the same management pathway: IV antibiotics and bowel rest. The prediction is a valid refinement of the gold, not a contradiction of it.

Action: 1.0

--- EXAMPLE B: CORRECT -- primary diagnosis covered, secondary missed ---

LLM diagnosis:

"Acute gallstone pancreatitis."

Ground truth:

"Gallstone pancreatitis
Choledocholithiasis"

Procedures: ["ERCP with sphincterotomy and stone removal", "Laparoscopic cholecystectomy"]

Thought: The gold lists two diagnoses; gallstone pancreatitis is first and therefore primary. The LLM correctly identifies it and misses the secondary (choledocholithiasis). Per principle 4, the primary being covered is sufficient. The procedures confirm surgical management occurred, consistent with gallstone pancreatitis. Importantly, procedures are not used to reclassify choledocholithiasis as primary -- the gold label ordering governs that.

Action: 1.0

--- EXAMPLE C: CORRECT -- extra diagnosis in same clinical domain ---

LLM diagnosis:

"Acute calculous cholecystitis with mild gallstone pancreatitis."

Ground truth:

"Acute cholecystitis"

Procedures: None documented.

Thought: The gold is acute cholecystitis. The LLM adds "mild gallstone pancreatitis" on top of the correct primary diagnosis. Both conditions arise from the same biliary pathology and are managed within the same clinical domain -- IV fluids, analgesia, NPO, and cholecystectomy planning. The added pancreatitis is a closely related complication, not a diagnosis that would redirect care to a fundamentally different pathway. Per principle 3, this is acceptable.

Action: 1.0

--- EXAMPLE D: INCORRECT -- related but distinct disease ---

LLM diagnosis:

"Symptomatic cholelithiasis (biliary colic)."

Ground truth:

"Acute cholecystitis"

Procedures: ["Laparoscopic cholecystectomy"]

Thought: Biliary colic and cholecystitis both involve gallstones but are distinct diagnoses with different urgency and management. Biliary colic is transient and managed expectantly; cholecystitis involves active gallbladder inflammation requiring urgent intervention, confirmed here by the cholecystectomy. Acting on biliary colic would delay necessary surgical care. Per principle 2, proximity on the gallstone spectrum does not make these equivalent.

Action: 0.0

--- EXAMPLE E: INCORRECT -- severity downgrade that changes management ---

LLM diagnosis:

"Acute diverticulitis."

Ground truth:

"Perforated diverticulitis"

Procedures: ["Exploratory laparotomy", "Sigmoid resection"]

Thought: The gold specifies perforation, a complication that mandates emergency surgery, as confirmed by the laparotomy and sigmoid resection. The LLM predicts uncomplicated diverticulitis, which would be managed with antibiotics alone. Per principle 1, when the gold specifies a severity that changes management, a general prediction is incorrect.

Action: 0.0

Now grade the following:

<begin LLM diagnosis>

{llm_diagnosis}

<end LLM diagnosis>

<begin ground truth diagnosis>

{ground_truth_diagnosis}

<end ground truth diagnosis>

<begin procedures performed>

{procedures}

<end procedures performed>

Use the Thought, Action paradigm. In your Thought, reason over whether the LLM diagnosis is clinically equivalent to the ground truth by asking: would a clinician acting on this prediction arrive at the same treatment decisions? Apply the principles above. If procedures are provided, use them as supporting evidence for severity and treatment intent, but treat them as potentially incomplete rather than definitive. If no procedures are listed, rely on clinical reasoning alone.

Thought: <insert thought here>

Then output an Action: 1.0 for correct, or 0.0 for incorrect.

Action: 0.0 or 1.0

Do not output anything after the Action line.

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Rare Disease Final Diagnosis Grader

We are given a medical diagnosis and a ground truth diagnosis.

We want to judge whether the diagnosis is correct.

This is actually a hard task for LLMs. Let me show you why.

The diagnosis:

"

Based on the provided information, the patient is a 47-year-old man with a history of cocaine addiction who was hospitalized after a cardiac arrest. He presented with post-anoxic encephalopathy, hyperpyrexia, disseminated intravascular coagulation (DIC), and multi-organ failure. The drug screening showed the presence of benzoylecgonine (a cocaine metabolite), ephedrine, MDA, MDMA, delta-9-tetrahydrocannabinol, and morphine.

The esophagogastroduodenoscopy revealed patchy areas of intense inflammation and necrosis in the gastric fundus, likely due to vasoconstriction and ischemia. The CT scan showed gastric intramural gas extending from the fundus along the greater curvature, which is indicative of emphysematous gastritis.

Emphysematous gastritis is a rare and severe form of gastritis characterized by the presence of gas within the wall of the stomach, often due to ischemia, infection, or a combination of factors, and is associated with a high mortality rate. In this patient, the combination of drug use, particularly cocaine, which is a potent vasoconstrictor, along with ephedrine, could have led to severe vasoconstriction and subsequent ischemia of the gastric wall, resulting in necrosis and gas formation.

Final Diagnosis: Emphysematous gastritis due to ischemia and necrosis from vasoconstriction associated with cocaine and ephedrine use.

"

And

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"

Based on these findings, a diagnosis of gastric cystic pneumatosis was made.

"

Are actually the same diagnosis. Or at least, they're close enough that physicians are satisfied.

However, consider this LLM diagnosis:

<begin LLM diagnosis>

Based on the information provided, the patient is a 34-year-old woman who presents with falls, weakness, new-onset jaundice, mild encephalopathy, and scleral icterus. The laboratory results indicate elevated liver enzymes, with a disproportionately higher AST compared to ALT, significantly elevated total bilirubin, and an increased INR.

This clinical picture suggests acute liver dysfunction with possible hepatic encephalopathy. The combination of neurological symptoms (falls, weakness, encephalopathy), jaundice, and the specific pattern of liver enzyme elevation (AST > ALT) is highly suggestive of Wilson's disease, particularly in a young adult. Wilson's disease is a genetic disorder leading to copper accumulation, which can cause liver damage and neurological symptoms.

Final Diagnosis: Wilson's disease

<end LLM diagnosis>

Meanwhile, the ground truth diagnosis is:

<begin ground truth diagnosis>

The patient was diagnosed with acute on chronic liver disease secondary to alcoholic hepatitis (AH).

<end ground truth diagnosis>

In this case, the LLM diagnosis is incorrect. And we need to mark it as 0.0 or serious harm may occur.

Grading with that sort of nuance is tricky.

Using these examples as your motivation, consider the following LLM diagnosis and ground truth diagnosis:

<begin LLM diagnosis>

{llm_diagnosis}

<end LLM diagnosis>

<begin ground truth diagnosis>

{ground_truth_diagnosis}

<end ground truth diagnosis>

Please use the Thought, Action paradigm to grade the LLM's diagnosis against the ground truth diagnosis.

First output a Thought where you reason over the LLM's diagnosis, and if it is close to the ground truth. Keep your Thought to under 50 words.

Thought: <insert thought here>

Then output an Action, which either 1.0 for correct, or 0.0 for incorrect.

Action: 0.0 or 1.0

Do not output anything else for the action, since we will rely on regex parsing downstream.

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Fact Matching Grader

You are a strict evaluator.

You are given:

1. ACTION_HISTORY: A chronological list of diagnostic actions and simulator results.
2. FACT_LIST: A list of target facts.

Task:

Determine which facts in FACT_LIST are supported by ACTION_HISTORY.

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```

Definitions:
- A fact counts as "discovered" ONLY IF the simulator RESULT explicitly states it,
  or directly entails it with no outside medical knowledge.

Rules:
- Do NOT use outside medical knowledge.
- Do NOT infer unstated facts.
- Do NOT invent facts not in FACT_LIST.
- Match facts semantically (paraphrases count; trivial formatting differences like 1200 vs 1,200 count)
.
- Count each fact at most once.
- Be conservative: if unsure, do NOT count.
- Important: Follow these rules strictly so the same rubric is applied to all runs. Same inputs must
yield the same list of indices.

Indexing rules (CRITICAL):
- FACT_LIST is indexed starting from 0.
- Indices refer to the exact position of each fact in FACT_LIST as provided.
- Do NOT renumber, reorder, or skip indices.
- Output indices in strictly increasing order.
- Each index may appear at most once.

Output format (EXACTLY ONE LINE, nothing else):
discovered_fact_indices: <JSON list of 0-based indices>

Rules for output:
- The list must be valid JSON.
- Indices must be integers.
- If no facts are discovered, output an empty list: []

-----
EXAMPLE (illustrative only)
-----

ACTION_HISTORY:
[
  {
    "action": "Order comprehensive metabolic panel",
    "result": "Lipase returned at 1200 U/L. ALT is 85 U/L."
  },
  {
    "action": "Ask about alcohol use",
    "result": "Patient denies alcohol use."
  }
]

FACT_LIST:
[
  "Lipase 1,200 U/L",
  "ALT 85 U/L",
  "AST 72 U/L",
  "No alcohol use"
]

Output:
discovered_fact_indices: [0,1,3]

-----
NOW EVALUATE THE FOLLOWING CASE
-----

ACTION_HISTORY:
{action_history}

FACT_LIST:
{fact_list}

```

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962 L.6 Baselines Diagnostic Loop Prompts

Action Prompt

You are given a patient's medical history.

Your job is to determine the next action that we should take
in trying to diagnose the patient as quickly as possible.

You should favor actions that will help you narrow down the diagnosis.

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Please keep in mind that doctors often (but not always) do physical exams.
 In the past, you've almost never done physical exams.
 Sometimes imaging is done before a physical exam, but not always.

It is also customary to only do a SINGLE ACTION at a time.
 For example, one physical exam, one set of medical imaging, one set of X-rays, etc.
 You are not allowed to ask for multiple actions at once. If you do, you will waste your turn and get a score of 0.0

Any underlines or blanks (e.g., "___", "____") in the medical history indicate intentionally redacted information.
 It is acceptable to base your decision on information that includes such redactions.

With all of that in mind, your job is to read the medical history below and propose an action.

```
<begin medical history>

{medical_history}

<end medical history>
```

Now please think of the next action that we should take to minimize the patient's time to diagnosis.

If you are ready to make a final diagnosis, please output
 Diagnosis: <insert diagnosis here>

Otherwise, please output the next action that we should take.

Next Action: <insert action here>

Our goal is to make a final diagnosis as quickly as possible.
 Your reward will be determined by how quickly you can make a diagnosis.
 However, if you guess incorrectly, you get zero reward. So please balance accordingly.
 A typical accuracy rate is around 60 percent for a skilled physician.

Output format rules (must follow exactly):

- If making a final diagnosis, output exactly one line:
 Diagnosis: <diagnosis>
- Otherwise, output exactly one line:
 Next Action: <action>

Do not include any other text, explanations, bullet points, or reasoning.

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Final Disease Prompt

Have you ever seen ImageNet? This model is evaluated by considering Top-K
 How accurate are the models K most likely image classes?

We want to consider a similar Top K scenario for the problem of medical diagnosis.

Specifically, you are given a medical report.
 You need to think about the medical report, and then output the top 3 most likely
 diseases or pathologies that are consistent with the data.
 In addition to the top 3 most likely diseases, you also need to output 2 diseases that are consistent
 with the data but less likely due to the rarity of the diagnosis.
 Think about this as sampling from the tails of the distribution. The diseases you pick should still be
 entirely plausible given the evidence. They should just be less likely because the underlying pathology
 is itself rare. Think Bayes rule.

Here is the medical report, which may be in progress

```
<begin medical report>

{medical_report}

<end medical report>
```

Your output should be a valid JSON string in the following format:

```
{
  "most_likely_diseases": [
    {
      "disease": "<insert disease one>",
      "rationale": "<your rationale here>"
    },
    {
      "disease": "<insert disease two>",
      "rationale": "<your rationale here>"
    }
  ]
}
```

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```

    }},
    {{
      "disease": "<insert disease three>",
      "rationale": "<your rationale here>"
    }}
  ],
  "rare_possibilities": [
    {{
      "disease": "<insert rare disease one>",
      "rationale": "<your rationale here>"
    }},
    {{
      "disease": "<insert rare disease two>",
      "rationale": "<your rationale here>"
    }}
  ]
}}

```

Please ensure the output is a properly formatted JSON string with no additional text before or after.

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967 L.7 GRPO Prompts

Rare Disease System Prompt

You are a medical diagnostic agent interacting with a simulator.

Your goal is to gather sufficient diagnostic information and then make a correct diagnosis. You must balance information gathering with timely decision-making.

At each step, output EXACTLY ONE LINE in one of the following two formats:

A: <one diagnostic action>

D: <one diagnosis label>

Rules:

- The first character of your output MUST be either "A" or "D".
- Do NOT output anything before "A:" or "D:".
- Do NOT explain your reasoning.
- Do NOT include multiple actions or diagnoses in one step.
- Do NOT ask questions directly to the patient.
- Actions must be diagnostic in nature, and must not be therapeutic or treatment-related.
- Do NOT repeat an action that has already been performed unless it is expected to yield new information.
- When you choose "A:", you will receive new information from the simulator, the result of the action.
- When you choose "D:", you are making your final diagnosis and the episode will end.

Guidelines:

- Use "A:" to gather information when the diagnosis is uncertain.
- Use "D:" only when you are sufficiently confident in the diagnosis.
- Prefer actions that reveal decisive or high-yield clinical information.
- Avoid unnecessary or repetitive actions.

You will be evaluated on:

- Diagnostic correctness
- Coverage of clinically important facts
- Appropriateness of when you decide to diagnose
- Efficiency (number of steps taken)

Output only the action or diagnosis line. Nothing else.

968

MIMIC-CDM System Prompt

You are a medical diagnostic agent interacting with a clinical simulator. Your goal is to gather diagnostic information through strategic actions and make a correct diagnosis.

How the Interaction Works

This is a multi-turn diagnostic exercise. On each turn:

1. Analyze the information you have so far
2. Decide whether to take another diagnostic action OR make your final diagnosis
3. Output exactly ONE line in the required format (see below)

Output Format - CRITICAL

Your output must be a single line in one of these two formats:

A: <one diagnostic action>

D: <one diagnosis label>

969

After you output an action, the simulator will provide results. You then analyze the updated information and output your next action or final diagnosis. This continues until you output a diagnosis, which ends the episode.

****Requirements:****

- Output ONLY a single line starting with "A:" or "D:"
- Do NOT add explanations, commentary, or additional text after the action/diagnosis
- Do NOT output multiple actions or diagnoses
- Do NOT output both an action and a diagnosis in the same turn

Guidelines for Diagnostic Actions (A:)

When you output an action, it should be:

- ****Diagnostic in nature****: Gather information to support diagnosis (not treatment)
- ****High-yield****: Likely to provide important clinical information
- ****Specific****: Be precise about what test or examination you want
- ****Non-repetitive****: Don't repeat actions already performed

Do NOT output:

- Therapeutic actions (medications, treatments, procedures)
- Direct questions to the patient (use examination or test orders instead)
- Vague requests (be specific about tests/examinations)

Note: After you output an action, any underlines or blanks (e.g., "___") in the simulator's results indicate intentionally redacted information.

Guidelines for Making a Diagnosis (D:)

When you output a diagnosis, the episode immediately ends. Only diagnose when:

- You have gathered sufficient information to support your conclusion
- You have reasonable confidence in your diagnosis
- Additional testing is unlikely to significantly change your assessment

Balance thoroughness (gathering adequate information) with timely decision-making.

Evaluation

You will be evaluated on:

- ****Diagnostic correctness****: Accuracy of your final diagnosis
- ****Information coverage****: Whether you gathered clinically important information
- ****Efficiency****: Number of steps taken to reach the correct diagnosis
- ****Timing****: Whether you diagnosed at an appropriate point

Output only the action or diagnosis line. Nothing else.

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974 the checklist: **The papers not including the checklist will be desk rejected.** The checklist should
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- 982 • Please provide a short (1–2 sentence) justification right after your answer (even for [N/A]).

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1001 Question: Do the main claims made in the abstract and introduction accurately reflect the
1002 paper’s contributions and scope?

1003 Answer: **[TODO]**

1004 Justification: **[TODO]**

1005 Guidelines:

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1009 contributions made in the paper and important assumptions and limitations. A [No] or
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1014 are not attained by the paper.

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1277 Guidelines:

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1285 collector.

1286 **15. Institutional review board (IRB) approvals or equivalent for research with human**
1287 **subjects**

1288 Question: Does the paper describe potential risks incurred by study participants, whether
1289 such risks were disclosed to the subjects, and whether Institutional Review Board (IRB)
1290 approvals (or an equivalent approval/review based on the requirements of your country or
1291 institution) were obtained?

1292 Answer: **[TODO]**

1293 Justification: **[TODO]**

1294 Guidelines:

1295 • The answer [N/A] means that the paper does not involve crowdsourcing nor research
1296 with human subjects.

1297 • Depending on the country in which research is conducted, IRB approval (or equivalent)
1298 may be required for any human subjects research. If you obtained IRB approval, you
1299 should clearly state this in the paper.

1300 • We recognize that the procedures for this may vary significantly between institutions
1301 and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the
1302 guidelines for their institution.

1303 • For initial submissions, do not include any information that would break anonymity (if
1304 applicable), such as the institution conducting the review.

1305 **16. Declaration of LLM usage**

1306 Question: Does the paper describe the usage of LLMs if it is an important, original, or
1307 non-standard component of the core methods in this research? Note that if the LLM is used
1308 only for writing, editing, or formatting purposes and does *not* impact the core methodology,
1309 scientific rigor, or originality of the research, declaration is not required.

1310 Answer: **[TODO]**

1311 Justification: **[TODO]**

1312 Guidelines:

1313 • The answer [N/A] means that the core method development in this research does not
1314 involve LLMs as any important, original, or non-standard components.

1315 • Please refer to our LLM policy in the NeurIPS handbook for what should or should not
1316 be described.